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Design and development of a vibration-damping hand-arm-exoskeleton

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Kurzfassung

Handgeführte Elektrowerkzeuge übertragen während des Betriebs mechanische Schwingungen unmittelbar auf Hand und Arm des Bedieners. Eine lang andauernde berufliche Exposition gegenüber solchen Schwingungen steht im Zusammenhang mit vaskulären, neurologischen und muskuloskelettalen Erkrankungen und kann zu dauerhaften Beeinträchtigungen der Handfunktion führen. Die Verringerung der auf die Hand einwirkenden Schwingungen ist daher für den Schutz von Beschäftigten, die regelmäßig vibrierende Werkzeuge bedienen, von großer Bedeutung. Herkömmliche am Körper getragene Schutzmaßnahmen beruhen überwiegend auf passiven Lösungen wie Antivibrationshandschuhen, deren Wirksamkeit von der Schwingungsfrequenz und der Wechselwirkung zwischen Hand und Werkzeug abhängt. Die aktive Schwingungsregelung verfolgt einen anderen Ansatz, indem sie eine Gegenschwingung erzeugt, die der auf die Hand einwirkenden Schwingung entgegenwirkt. Die vorliegende Arbeit beschreibt die systematische Entwicklung und experimentelle Validierung eines tragbaren Handexoskeletts zur aktiven Schwingungsregelung, das die von einem handgeführten Elektrowerkzeug auf den Anwender übertragenen Schwingungen reduzieren soll.

Die Entwicklung des Handexoskeletts erfolgt nach der Richtlinie VDI 2221 und führt zu einem funktionsfähigen, am Körper tragbaren Prototyp für die Schwingungsregelung an der Hand. Das System umfasst einen Schwingungssensor am Werkzeug zur Erfassung der eingeleiteten Schwingung, einen zweiten Sensor auf dem Handrücken zur Erfassung der beim Anwender ankommenden Schwingung sowie einen elektrodynamischen Aktor am Handgelenk zur Erzeugung der Gegenschwingung. Die Messsignale werden von einem Mikrocontroller mit einem eigens entwickelten adaptiven Regelalgorithmus verarbeitet. Auf Grundlage des Filtered-x-Least-Mean-Squares-Verfahrens (FxLMS) identifiziert der Regler die dominante Werkzeugschwingung, erlernt, wie die vom Aktor erzeugte Schwingung an die Hand übertragen wird, und passt die Gegenschwingung während des Betriebs selbstständig so an, dass sie der gemessenen Schwingung entgegenwirkt.

Die Leistungsfähigkeit der Schwingungsregelung des entwickelten Systems wird experimentell mit drei gesunden Teilnehmenden beim Bedienen eines handgeführten Elektrowerkzeugs anhand von Versuchen zum adaptiven Lernen, zur Wiederholbarkeit, zur anhaltenden Regelung sowie zum Aus- und Einschalten der Regelung untersucht. Dieselben drei Teilnehmenden nahmen zusätzlich an einer strukturierten Befragung zur Bewertung der Erfahrungen mit dem tragbaren System teil.

Die Ergebnisse zeigen, dass das System selbstständig eine wirksame Gegenschwingung erlernt und in den anschließenden Validierungsversuchen die dominante Schwingung an der Hand in der geregelten Richtung um bis zu etwa 60 % reduziert. Das erlernte

Kompensationsverhalten ist in wiederholten Versuchen reproduzierbar und bleibt auch ohne weitere Adaption wirksam, während die Rückmeldungen der Teilnehmenden darauf hinweisen, dass die Schwingungsminderung während der Werkzeugbedienung wahrnehmbar ist. Insgesamt zeigen diese Ergebnisse die Machbarkeit einer tragbaren aktiven Schwingungskompensation unmittelbar an der Hand und bilden eine solide Grundlage für die Weiterentwicklung eines mehrachsigen aktiven Schwingungsschutzes für industrielle Anwendungen mit Handwerkzeugen.

Abstract

Powered hand tools transmit mechanical vibration directly into the operator's hand and arm during use. Prolonged occupational exposure to such vibration is associated with vascular, neurological and musculoskeletal disorders and can lead to persistent impairment of hand function. Reducing the vibration reaching the hand is therefore important for protecting workers who regularly operate vibrating tools. Conventional wearable protection relies mainly on passive measures such as anti-vibration gloves, whose effectiveness depends on vibration frequency and the interaction between the hand and tool. Active vibration control provides a different approach by generating an opposing vibration that counteracts the vibration reaching the hand. This thesis presents the systematic development and experimental validation of a wearable active vibration-control hand exoskeleton designed to reduce vibration transmitted from a powered hand tool to the user.

The development of the hand exoskeleton follows the VDI 2221 guidelines, resulting in a functional wearable prototype designed for vibration control at the hand. The system incorporates a vibration sensor on the tool to measure the incoming vibration, a second sensor on the back of the hand to measure the vibration experienced by the user, and an electrodynamic actuator at the wrist to generate the counter-vibration. The measured signals are processed by a microcontroller running a custom adaptive control algorithm. Based on the filtered-x least mean squares (FxLMS) method, the controller identifies the dominant tool vibration, learns how the actuator-generated vibration reaches the hand, and automatically adjusts the counter-vibration during operation to oppose the measured vibration.

The vibration-control performance of the developed system is evaluated experimentally with three healthy participants operating a handheld power tool through adaptive learning, repeatability, sustained-control and OFF–ON switching tests. The same three participants also completed a structured questionnaire assessing the wearable experience.

The results show that the system automatically learns an effective counter-vibration and subsequently achieves up to approximately 60 % reduction of the dominant vibration at the hand in the controlled direction during validation. The learned cancellation behaviour is reproduced across repeated tests and remains effective when applied without further adaptation, while the participant feedback indicates that the vibration reduction is perceptible during tool operation. Together, these outcomes demonstrate the feasibility of wearable active vibration cancellation directly at the hand and provide a strong foundation for the further development of multi-directional active vibration protection for industrial hand-tool applications.

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List of Abbreviations

BC	Basicentric
BD	Biodynamic
DAC	Digital-to-analogue converter
DC–DC	Direct-current-to-direct-current converter
DFT	Discrete Fourier transform
DIN	Deutsches Institut für Normung (German Institute for Standardization)
EN	European Standard (Europäische Norm)
FFT	Fast Fourier transform
FxLMS	Filtered-x least-mean-squares
ISO	International Organization for Standardization
LMS	Least-mean-squares
LRA	Linear resonant actuator
MEMS	Microelectromechanical systems
NCO	Numerically controlled oscillator
PC	Personal computer
PID	Proportional–integral–derivative
SPI	Serial Peripheral Interface
USB	Universal Serial Bus
VDI	Verein Deutscher Ingenieure (Association of German Engineers)

1 Introduction

This chapter presents the motivation for developing an active hand exoskeleton for reducing occupational hand–arm vibration, highlighting the need for improved wearable protection during powered hand-tool operation. Following this, the central problem is defined by considering the limitations of existing passive vibration-reduction measures and the potential of active vibration control. Finally, the objectives of this thesis are introduced, establishing the basis for the systematic development and experimental evaluation of the prototype. An overview of the thesis structure is also provided.

1.1 Motivation

Powered hand tools are widely used in manufacturing, construction, maintenance and other industrial activities. During operation, mechanical vibration is transferred from the tool through the hand into the upper limb. Repeated occupational exposure to hand-transmitted vibration is associated with vascular, neurological and musculoskeletal disorders and can lead to persistent impairment of hand function [Bov98; Nil17]. Reducing the vibration experienced by the operator is therefore an important part of protecting workers who regularly use vibrating equipment.

Where vibration cannot be sufficiently reduced at its source, wearable protection provides an additional means of limiting exposure. Anti-vibration gloves are the most established example, but their effectiveness depends strongly on vibration frequency and the mechanical interaction between the hand and tool [Don04]. Their performance can also differ considerably across the hand: experimental measurements have shown that attenuation at the palm does not necessarily correspond to attenuation at the fingers, where vibration amplification can occur at some frequencies [Ham18].

These limitations create an opportunity to investigate vibration-reduction approaches that are not governed only by fixed material and structural properties. Active vibration control uses powered actuation to generate a controlled mechanical response intended to oppose unwanted vibration [Ell87]. This provides greater freedom to modify the response during operation, but also introduces additional requirements for vibration measurement, actuation, control and mechanical integration.

Applying such an approach directly at the hand creates a distinct wearable engineering challenge. The system must interact mechanically with the user while preserving the ability to grip and operate the tool, and its sensing, actuation and control functions must operate together within the wearable arrangement. Existing experimentally validated systems for mitigating industrial hand–arm vibration are predominantly based on passive isolation, damping or altered load transfer, whereas active vibration reduction applied directly through a compact hand-worn system remains less extensively investigated

[Ciu21; Raj21]. The present work therefore investigates a wearable active vibration-control system for powered hand-tool operation.

1.2 Problem Statement

Powered hand tools can expose the operator to vibration that is transmitted directly into the hand and arm. Existing wearable protection is based mainly on passive isolation and damping. Although these measures can reduce vibration under favourable conditions, their effectiveness varies with the excitation and the mechanical interaction between the user and the tool. This motivates the investigation of a wearable solution that can respond actively to vibration during operation rather than relying only on fixed passive properties. Active vibration control provides this possibility by generating a controlled mechanical response that counteracts the incoming vibration. The central problem, therefore, is the challenge of engineering a wearable active vibration-control system that can measure the vibration produced during tool operation, generate an appropriate opposing response and adapt that response so that the vibration reaching the hand is reduced.

1.3 Objectives

The primary objective of this master's thesis is to develop and experimentally evaluate a functional prototype of a wearable active vibration-control hand exoskeleton for reducing vibration transmitted from powered hand tools to the user.

The specific objectives of this work are to:

- design the functional architecture and wearable structure of the active vibration-control system using a systematic product-development methodology;
- develop and assemble a functional prototype integrating the required vibration sensing, actuation, control and supporting hardware;
- develop and implement an adaptive control strategy capable of determining and adjusting the counter-vibration required during tool operation; and
- experimentally evaluate the prototype with respect to vibration reduction, repeatability, continued operation and wearable usability.

1.4 Structure of the Thesis

- Chapter 1: Introduction – Presents the motivation, problem statement, objectives and structure of the thesis.
- Chapter 2: State of the Art – Reviews hand–arm vibration, existing mitigation approaches and the sensing, actuation, control and product-development principles relevant to wearable active vibration reduction.
- Chapter 3: Methodology – Describes the systematic development of the prototype, including concept generation, hardware integration, vibration measurement, control development and the experimental procedure.
- Chapter 4: Results and Discussion – Presents and discusses the experimental performance of the developed system, including vibration reduction, repeatability, sustained control behaviour and wearable evaluation.

-
- Chapter 5: Summary and Outlook – Summarises the main findings and identifies the principal directions for further development.

2 State of the Art

This chapter establishes the scientific and engineering basis for the development of a wearable active vibration-control system. It first considers the occupational relevance and characteristics of hand–arm vibration and the way vibration is transmitted through the coupled tool hand arm system. The available mitigation approaches are then compared, establishing the motivation for active intervention and the actuation, control, measurement and signal-processing technologies required to realise it. The available evidence is subsequently synthesised to identify the remaining research gap, after which systematic product-development methods are considered as the framework for translating the resulting requirements into a technical concept.

2.1 Hand–Arm Vibration and Occupational Exposure

Hand–arm vibration describes mechanical vibration transmitted from a tool, workpiece or machine through the hand into the upper limb. Its occupational significance depends not only on vibration magnitude, but also on frequency content, exposure duration and the conditions under which the tool is operated. Understanding the associated health effects, the methods used to quantify occupational exposure and the vibration characteristics produced by industrial tools therefore provides the basis for evaluating measures intended to reduce that exposure.

2.1.1 Hand–Arm Vibration and Health Effects

Long term exposure to hand-transmitted vibration is associated with a group of disorders collectively described as hand–arm vibration syndrome. Figure 1 summarises how exposure to hand-transmitted vibration is linked to its effects on the human body [Don21].

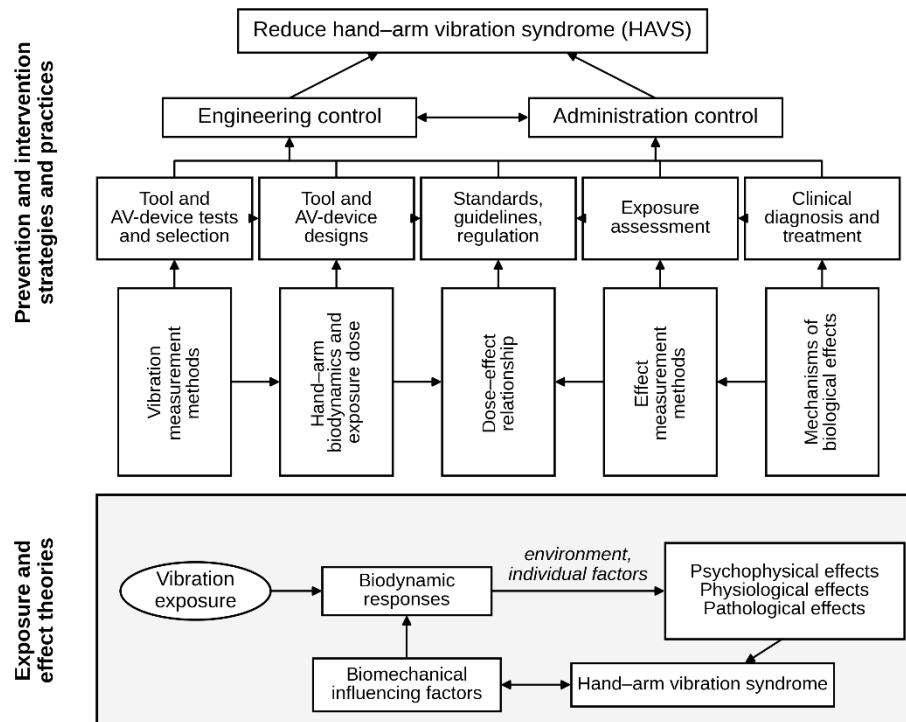


Figure 1: Hand-transmitted vibration exposure and its health effects [Don21].

The associated disorders are generally grouped into vascular, neurological and musculoskeletal effects [Bov98]. Vascular effects impair peripheral blood circulation in the fingers and can lead to vibration-induced finger blanching, in which the fingers turn white during vasospastic attacks [Bov98; Nil17]. Neurological effects include numbness, tingling and reduced tactile sensitivity, which can impair the ability to perceive and manipulate objects [Bov98]. Musculoskeletal effects involve the bones, joints and muscles of the hand and arm [Bov98]. These disorders may develop gradually with continued exposure and, in advanced stages, can result in persistent impairment of hand function [Bov98]. Epidemiological evidence confirms a clearly increased risk of vascular and neurological disorders among workers exposed to occupational hand–arm vibration [Nil17]. Because occupational exposure depends on both the magnitude of the vibration and the duration for which it acts on the worker, these factors must be combined in a consistent way to allow meaningful assessment and comparison [ISO53491].

2.1.2 Occupational Exposure and Regulatory Framework

DIN EN ISO 5349-1 provides the basis for the measurement and evaluation of hand-transmitted vibration, while DIN EN ISO 5349-2 gives practical guidance for workplace measurements [ISO53491; ISO53492]. The requirements for the measuring instruments are defined in ISO 8041-1 [ISO8041]. Vibration is measured in three orthogonal directions and frequency-weighted, after which the weighted components are combined into a vibration total value. The duration of exposure is then incorporated by normalising the result to an eight-hour reference period and expressing it as the daily vibration exposure $A(8)$ [ISO53491]. The weighting itself remains subject to research, since not all weighting

approaches represent the relationship between vibration frequency and vascular risk in the same way [Bov12].

On this basis, Directive 2002/44/EC defines a daily exposure action value of $2.5 \text{ m/s}^2 A(8)$ above which measures to reduce exposure are required, and an exposure limit value of $5.0 \text{ m/s}^2 A(8)$, which must not be exceeded [EU2002/44]. The practical relationship between vibration magnitude and the exposure duration required to reach these thresholds is shown in Figure 2.

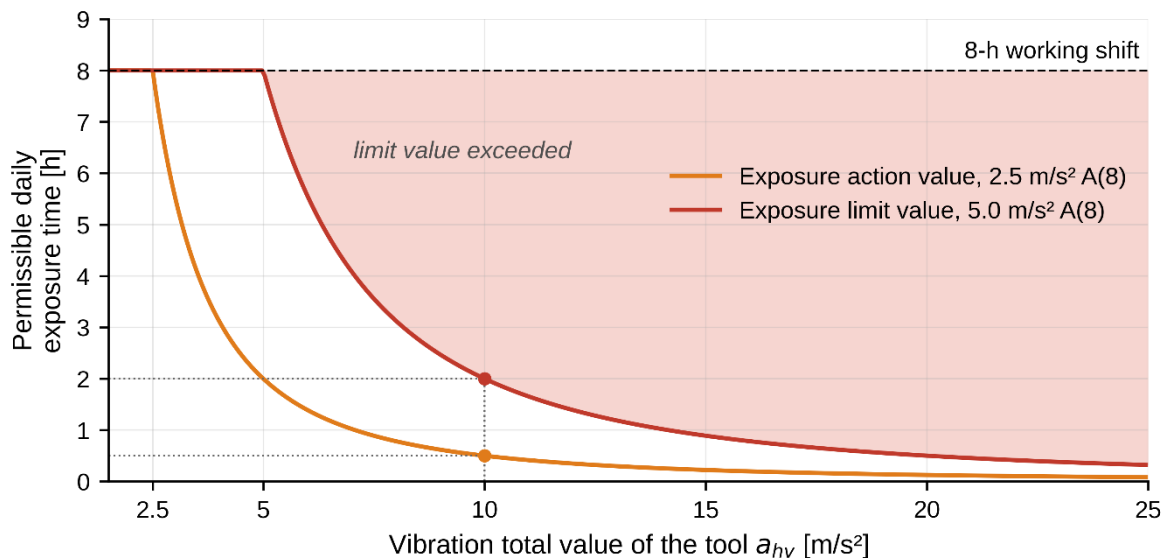


Figure 2: Daily exposure time until the action and limit values are reached as a function of the vibration total value [ISO53491; EU2002/44].

The allowable exposure duration falls rapidly as vibration magnitude increases. A vibration total value of 10 m/s^2 , for example, reaches the action value after approximately 30 minutes and the exposure limit after approximately two hours of daily use. For strongly vibrating tools, relevant exposure levels can therefore be reached well before the end of a normal working shift. Reducing exposure without relying only on shorter operating times consequently requires reducing the vibration transmitted to the worker, and the effectiveness of such measures depends on the characteristics of the vibration generated during tool operation.

2.1.3 Vibration Characteristics of Industrial Hand Tools

Industrial hand tools generate vibration patterns that depend on the excitation mechanism, tool configuration and operating condition [Mon23; Bra26]. Figure 3 contrasts the characteristic time- and frequency-domain responses of rotating, impact-driven and combined tool mechanisms.

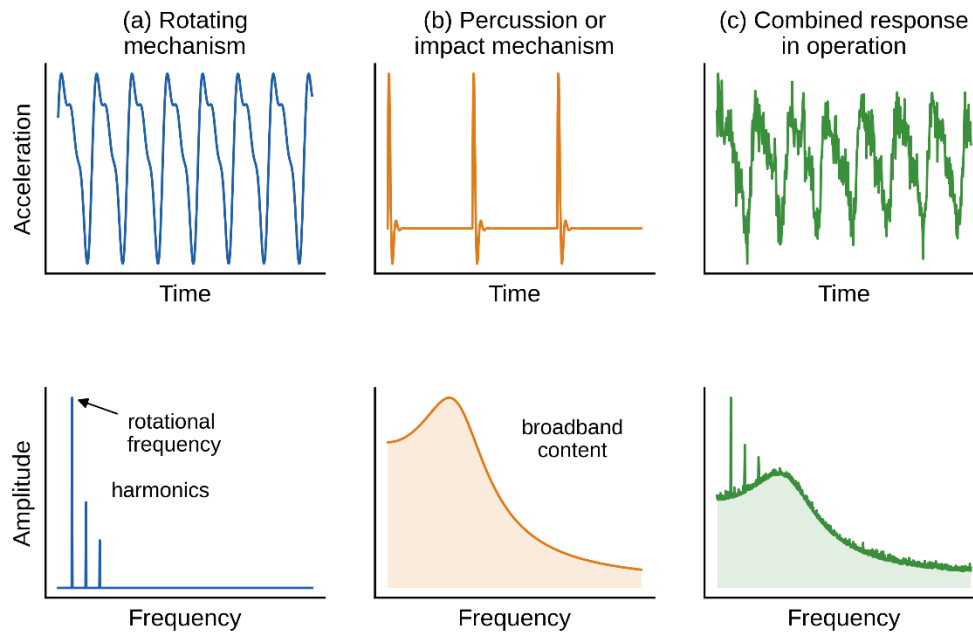


Figure 3: Schematic vibration signatures of (a) rotating, (b) impact-driven and (c) combined tool mechanisms in the time and frequency domains, based on [Mon23; Bra26].

Rotating excitation produces comparatively regular periodic content with distinct spectral components, whereas repeated impacts introduce short-duration events with energy distributed over a broader frequency range. During practical tool operation, these mechanisms can occur together, so the vibration entering the hand may contain periodic, broadband and transient components rather than a single fixed frequency.

Manufacturer-declared vibration emission values provide standardised information for comparing tools, but they do not necessarily represent the vibration produced during a specific working task. Measurements under simulated workplace conditions have shown clear differences between declared and task-related vibration values [Rim08]. The vibration relevant to a particular application must therefore be characterised under representative operating conditions.

The tool determines the excitation available at the contact interface, but the vibration ultimately experienced by the user also depends on the mechanical interaction between the tool, hand and arm.

2.2 Tool Hand Transmission and Biodynamic Response

The vibration generated by a tool is not transmitted unchanged into the hand and arm. Instead, the tool, contact interface and upper limb form a coupled mechanical system whose response depends on both the excitation and the interaction between the user and the tool [Don21; Xu21]. This response is governed by hand–tool coupling, biodynamic behaviour, frequency, direction and measurement location.

2.2.1 Hand–Tool Coupling and Vibration Transmission

Mechanical coupling between the hand and the vibrating source determines how vibration is transferred across the contact interface. Grip force, push or feed force and hand–arm

posture alters this coupling and can consequently modify the vibration transmitted to the user [Don21; Xu21]. Measurements with powered hand tools have also shown that changes in gripping and feed force influence the vibration response measured during operation [Lin21].

Figure 4 demonstrates this effect using vibration transmissibility measured at the hand dorsum, wrist, forearm and upper arm for two different feed forces [Xu21].

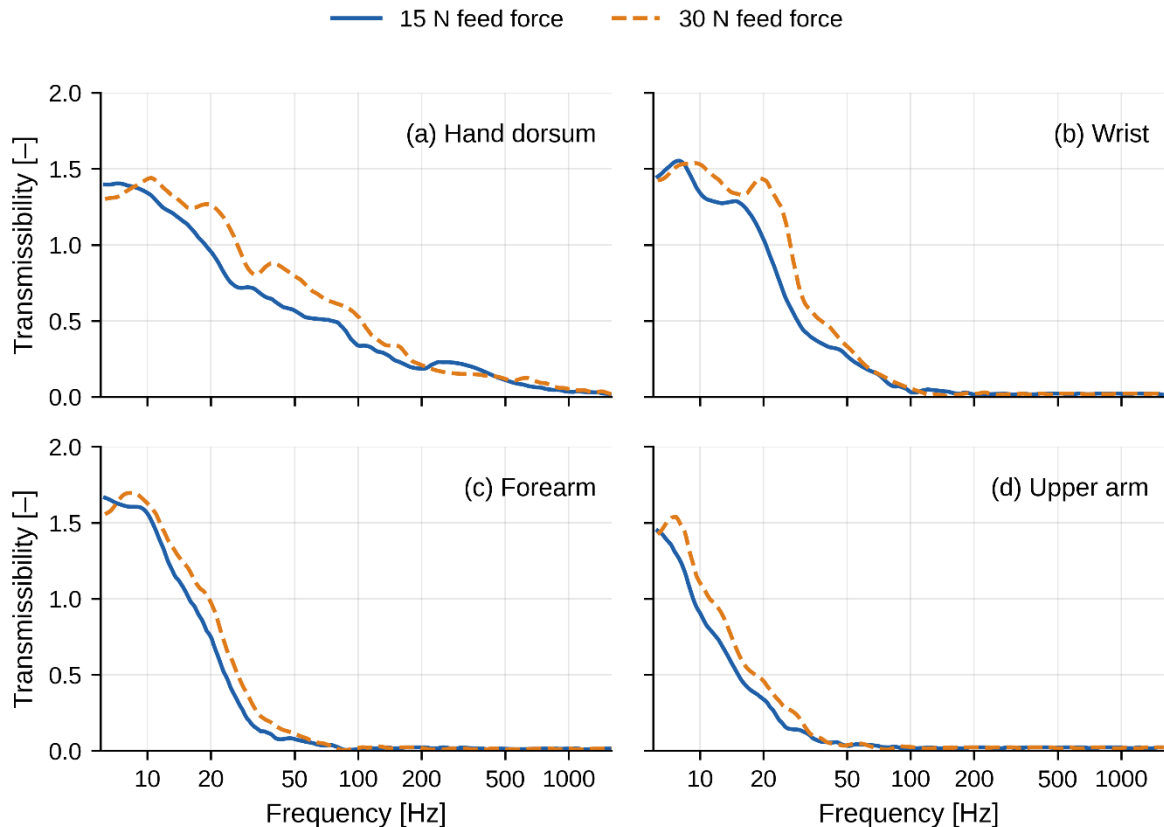


Figure 4: Median vibration transmissibility at (a) the hand dorsum, (b) wrist, (c) forearm and (d) upper arm for feed forces of 15 N and 30 N [Xu21].

The measurements in Figure 4 show that increasing the feed force from 15 N to 30 N increases vibration transmission at the measured locations [Xu21]. The response also decreases as the measurement position moves away from the point of excitation, demonstrating that vibration transmission depends simultaneously on the mechanical coupling at the hand and on the position within the upper limb [Xu21].

The hand–arm system therefore cannot be considered a rigid receiver of tool vibration. Its own mechanical properties influence the relationship between the excitation applied at the contact interface and the motion produced throughout the upper limb [Don21].

2.2.2 Biodynamic Response of the Hand–Arm System

The interaction between the vibrating source and the hand–arm system can be characterised using quantities such as mechanical impedance, apparent mass and vibration transmissibility [Don21]. Mechanical impedance relates the applied force to the resulting vibration velocity, while apparent mass relates force to acceleration.

Transmissibility describes how the vibration response changes between two measurement locations. Reference values for the driving-point mechanical impedance of the hand–arm system under defined excitation conditions are provided in ISO 10068 [ISO10068].

Figure 5 shows the apparent mass and mechanical impedance of the hand–arm system measured for two different feed forces [Xu21].

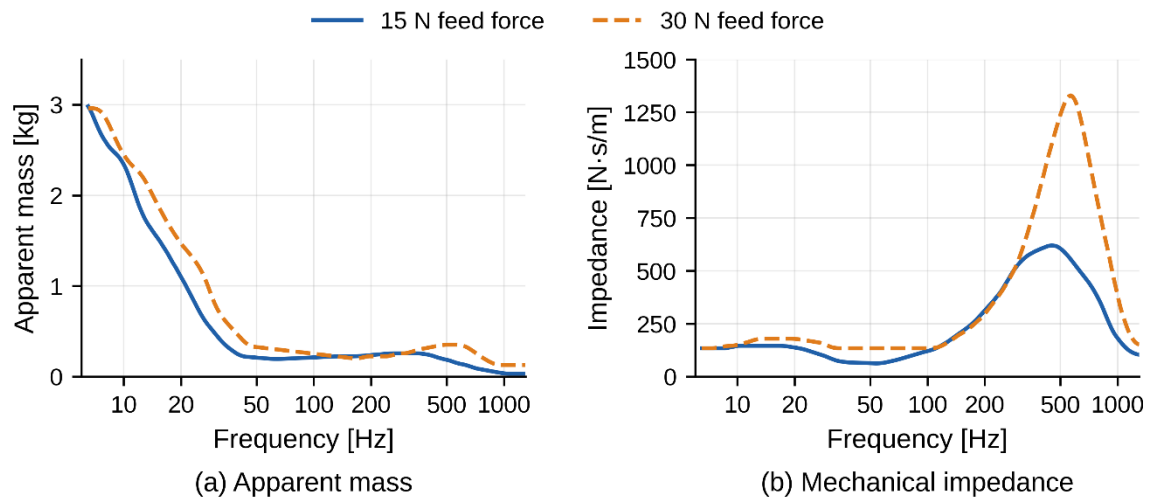


Figure 5: Driving-point (a) apparent mass and (b) mechanical impedance of the hand–arm system for feed forces of 15 N and 30 N [Xu21].

Figure 5 shows that the hand–arm system does not behave as a single rigid mass. Its apparent mass decreases strongly with increasing frequency, while the mechanical impedance exhibits a pronounced peak at several hundred hertz. Increasing the feed force raises the impedance peak considerably, indicating that the dynamic response depends on how firmly the tool is held [Xu21].

2.2.3 Frequency, Direction and Measurement Location

The frequency of excitation strongly influences how vibration propagates through the hand–arm system. Measurements in Figure 4 show that transmission decreases markedly as frequency increases and as the measurement location moves farther from the vibration source [Xu21]. In the reported tests, less than approximately 30% of the input vibration reached the hand dorsum above 160 Hz, while transmission to the wrist, forearm and upper arm fell below approximately 5% above 100 Hz, 63 Hz and 40 Hz, respectively [Xu21].

The response also depends on vibration direction because the anatomical and mechanical properties of the hand–arm system differ between excitation axes [Don21; ISO10068]. Directional measurements must therefore be related to a defined coordinate system; the two conventions used for hand-transmitted vibration are shown in Figure 6.

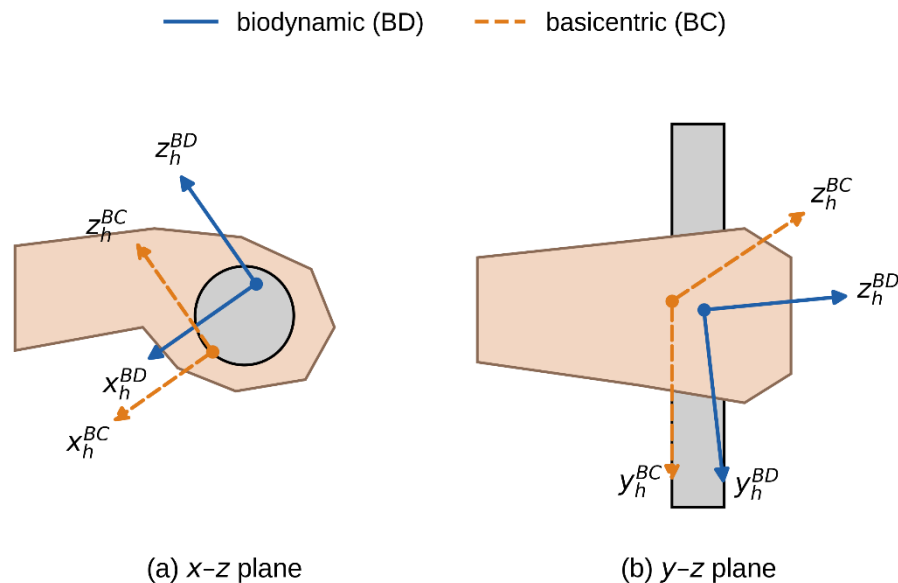


Figure 6: Biodynamic (BD) and basicentric (BC) hand coordinate systems in (a) the x-z plane and (b) the y-z plane [Don15; ISO53491].

The biodynamic and basicentric systems provide different reference conventions for expressing vibration direction relative to the hand and the hand–handle interface. Specifying the coordinate system is therefore important when directional vibration measurements and biodynamic responses are compared [Don15; ISO53491].

Frequency, direction and measurement location must therefore be considered together when describing vibration transmission through the upper limb [Don21; Xu21; ISO10068]. A vibration-reduction measure can consequently be effective only if its mechanical behaviour is compatible with the way vibration is transmitted under the relevant operating conditions.

2.3 Vibration Mitigation Strategies

Vibration transmitted through the hand–arm system can be reduced by modifying the mechanical transmission path or by introducing an additional controlled mechanical response [Hew16; Ben04]. Passive systems rely on fixed mechanical properties, semiactive systems vary properties such as stiffness or damping during operation and active systems use powered actuation to generate a controlled force [Spe97; Ben04]. These approaches therefore differ mainly in their adaptability, control authority and technical complexity.

2.3.1 Passive Vibration Mitigation

Passive vibration mitigation reduces transmission through fixed mechanical properties such as compliance and damping without requiring external power for control [Hew16]. In hand–arm applications, common examples include vibration-reducing gloves and mechanically isolated or suspended handles [Don04; Odd04].

Vibration-reducing gloves introduce a compliant layer between the handle and the hand [Don04]. Figure 7 compares middle-finger vibration transmissibility for several glove constructions [Ham18].

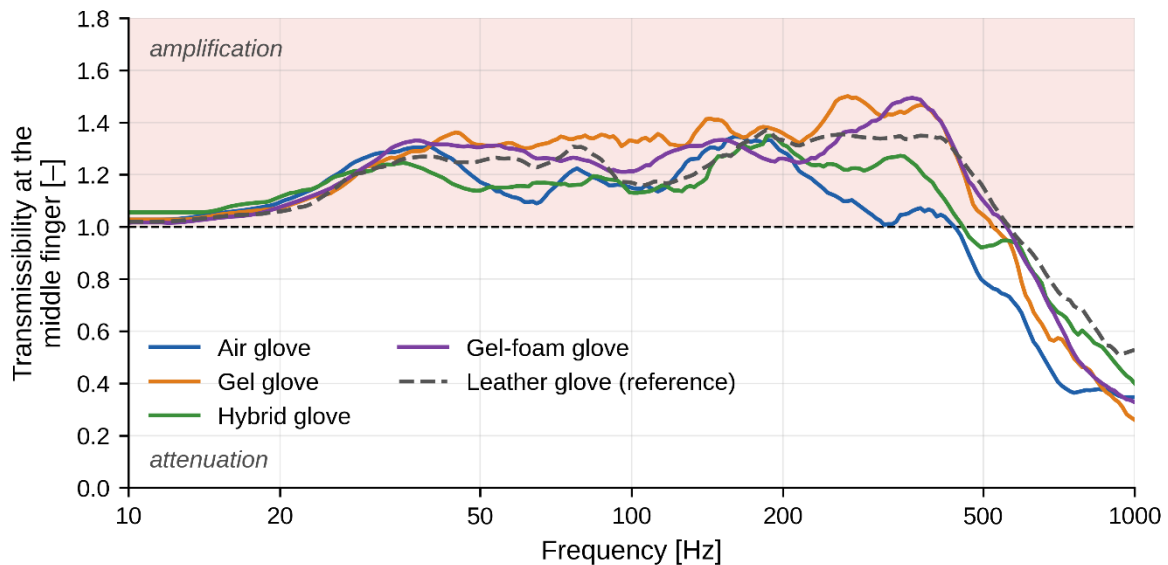


Figure 7: Mean vibration transmissibility at the middle finger for different vibration-reducing glove constructions [Ham18].

The measured attenuation is not uniform across frequency. Transmissibility exceeds unity at some frequencies, indicating vibration amplification, while attenuation occurs over other frequency ranges and differs between glove constructions [Ham18]. Suspended handles similarly depend on the mechanical relationship between the suspension and the transmitted vibration [Odd04].

Passive systems can therefore provide effective attenuation under suitable conditions, but their response remains governed by fixed mechanical properties and can change when the excitation or coupling condition changes [Hew16; Don04]. Greater adaptability requires these properties to be adjustable during operation [Spe97].

2.3.2 Semiactive Vibration Mitigation

Semiactive systems retain a passive mechanical transmission path but allow selected properties such as stiffness or damping to be varied during operation [Spe97]. Magnetorheological elements are one example because their mechanical resistance can be altered by changing the applied magnetic field [Spe97].

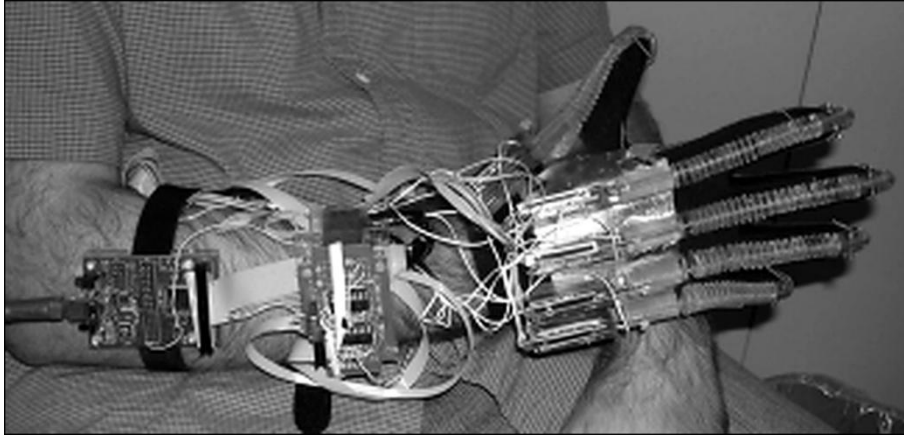


Figure 8: MRAGES force-feedback glove with magnetorheological actuators [Win07].

A wearable example is the MRAGES force-feedback glove, in which magnetorheological actuators provide controllable mechanical resistance at the fingers [Win07] (Figure 8).

Unlike the active-cancellation concept investigated in this thesis, the MRAGES system changes the resistance of the mechanical path rather than independently generating a counter-vibration at the hand [Win07; Spe97].

Semiactive approaches therefore provide greater adaptability than fixed passive elements, but their influence remains limited by the range over which the mechanical properties can be changed [Spe97]. Generating an additional mechanical response independently of these passive properties requires active actuation [Ben04].

2.3.3 Active Vibration Mitigation

Active vibration mitigation uses external power to drive an actuator that generates a controlled mechanical action directly on the vibrating system [Ben04]. For periodic or narrowband disturbances, the generated response can be adjusted in amplitude and phase so that it opposes the unwanted vibration at the controlled location [Liu18]. The effectiveness of active control therefore depends not only on the actuator itself, but also on the control strategy and the mechanical dynamics between the actuator and the controlled structure [Ben04]. Active concepts have also been investigated experimentally in powered hand-tool applications using actively suspended handles, demonstrating that controlled actuation can be introduced directly into the tool–hand vibration path [Maz17].

2.3.4 Rationale for Active Investigation

Active control is therefore relevant where the mechanical response must be adjusted during operation rather than being fixed by passive properties. For a wearable implementation, the next engineering question is which actuator can provide the required response within the available size, mass and integration constraints [Ben04].

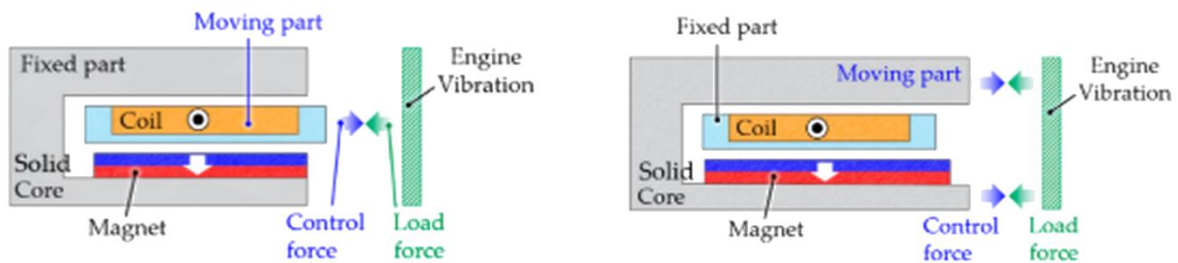
2.4 Actuation Technologies for Active Vibration Control

The actuator converts the electrical control command into a mechanical response. For a wearable system, candidate technologies must be considered in terms of force capability,

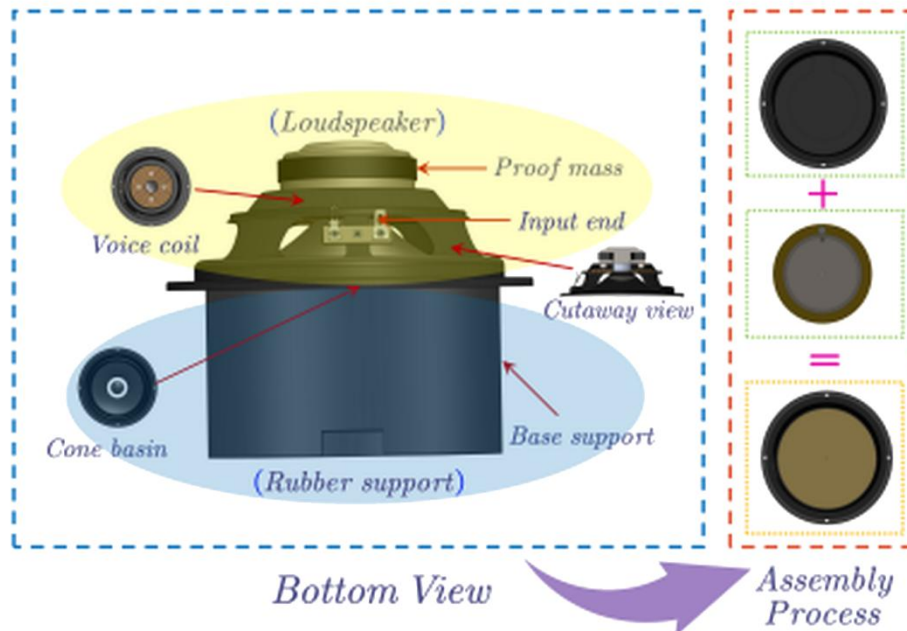
usable frequency range, displacement, dynamic response, mass, power demand and mechanical integration [Ben04]. Their installed response also depends on the coupled mechanical structure.

2.4.1 Electrodynamic and Voice Coil Actuators

Electrodynamic actuators generate mechanical force through the interaction between an electrical current and a magnetic field. Voice-coil actuators provide continuously controllable motion, while inertial-mass configurations generate a reaction force through the motion of an internal mass [Che05; Ben04]. Compact surface exciters apply the same principle within a small package and can therefore be attached directly to a structure to inject vibration [Che26; Mao26]. Figure 9 compares conventional voice-coil and inertial-mass arrangements with a compact loudspeaker-based inertial actuator.



(a) Voice-coil actuator (left) and inertial-mass linear actuator (right)



(b) Loudspeaker-based electrodynamic actuator

Figure 9: Electrodynamic actuator concepts: (a) voice-coil and inertial-mass linear actuators [Hon23]; (b) loudspeaker-based inertial actuator [Che23b].

All three arrangements use electromagnetic interaction to generate mechanical motion, but they differ in which element moves and how the resulting force is transferred. The

compact inertial configuration is particularly relevant for vibration-control applications because it can be attached directly to a structure while providing continuously controllable mechanical excitation.

2.4.2 Linear Resonant Actuators

Linear resonant actuators (LRAs) are compact electromagnetic actuators in which a spring-supported moving mass oscillates predominantly along a defined direction [Sch20]. The moving mass and suspension form a resonant mechanical system, allowing efficient vibration generation near the actuator's natural frequency [Sch20; Liu23]. Figure 10 shows both the internal construction of a compact LRA and an example of wearable integration.

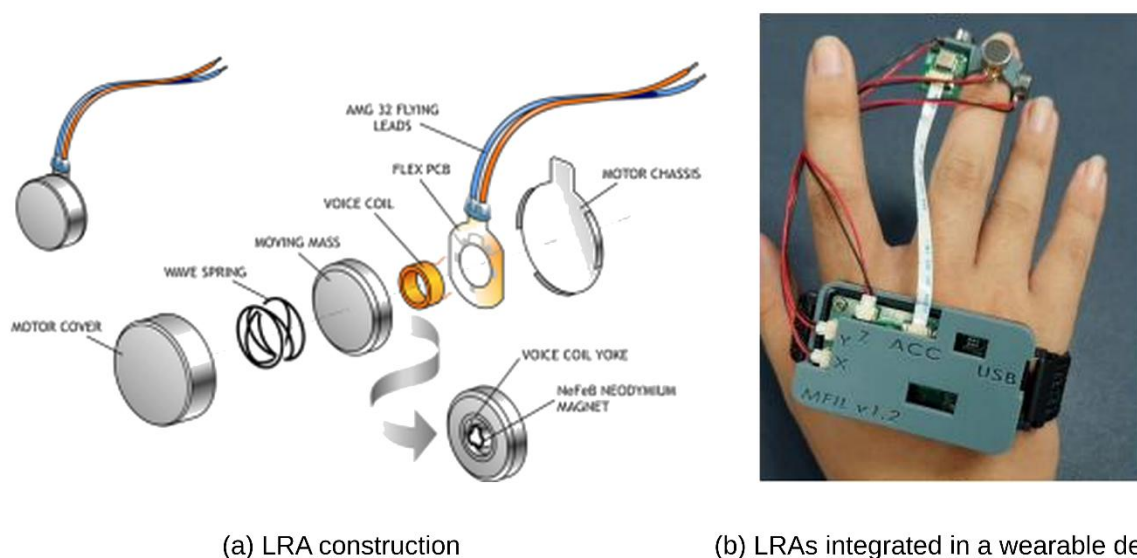


Figure 10: Linear resonant actuators: (a) construction of a compact LRA; (b) LRAs integrated in a wearable device [Liu23].

The wearable implementation demonstrates the principal integration advantage of LRAs: several compact actuators can be positioned directly on a user-mounted device [Liu23]. Their resonant mechanical design, however, restricts the frequency range over which a strong response can be generated. LRAs are therefore particularly suitable when compactness and efficiency are important and the required operating frequency lies close to the actuator resonance [Sch20; Liu23].

2.4.3 Piezoelectric Actuators

Piezoelectric actuators generate mechanical deformation when an electric field is applied to a piezoelectric material [Shi20]. They provide fast response, high stiffness and substantial force capability within a compact form, although the available displacement is generally limited and dedicated drive electronics may be required [Shi20]. Piezoelectric stacks have also been applied in active suspended hand-tool handles, where their high force and rapid response were suitable for vibration control, while actuator stroke and saturation constrained the usable operating range [Maz17]. Piezoelectric actuation

therefore offers high force capability and rapid dynamic response, but with more restrictive displacement and drive requirements than electrodynamic alternatives [Shi20; Maz17].

2.4.4 Alternative Actuation Technologies

Pneumatic actuators provide another means of generating controllable mechanical forces through regulated air pressure. Active vibration-isolation systems using air springs and servo valves have demonstrated adjustable vibration-control behaviour [Kat07].

Their use in a compact wearable device, however, requires additional components such as valves, regulators and a pressurised air supply [Kat07]. These supporting requirements make pneumatic systems more difficult to integrate into a self-contained hand-worn arrangement.

The actuator technologies considered therefore involve different compromises between force, displacement, frequency response, compactness and supporting hardware [Ben04; Shi20]. Their suitability depends on matching these characteristics to the required mechanical response and wearable constraints.

2.5 Control Strategies for Active Vibration Reduction

Once an actuator can generate the required mechanical response, the control strategy determines how that response is applied during operation. Depending on the available information, the actuator command can be prescribed in advance, determined from the measured response, or adapted using a reference signal related to the disturbance [Ell87; Liu17].

2.5.1 Prescribed Harmonic (Open Loop) Control

When vibration is dominated by a stable periodic component, the actuator can be driven with a harmonic signal at the same frequency [Liu17]. Effective cancellation requires an appropriate amplitude and phase relationship between the disturbance and the actuator-generated response at the controlled location [Ell87; Liu17].

This approach is comparatively simple because the actuator command can be defined directly once the required frequency, amplitude and phase are known [Liu17]. Its effectiveness decreases, however, if the disturbance frequency or the required phase relationship changes during operation, since the predefined command does not adjust itself to the resulting vibration [Liu17].

Using the measured vibration response to modify the command provides a means of correcting such changes during operation [Ell87].

2.5.2 Feedback Control

Feedback control uses the measured response at the controlled location to determine the actuator command. Deviations in the vibration response are returned to the controller,

which continuously modifies the control action according to the measured output [Ben04; Raj21]. A representative hand–glove feedback architecture is shown in Figure 11 [Raj21].

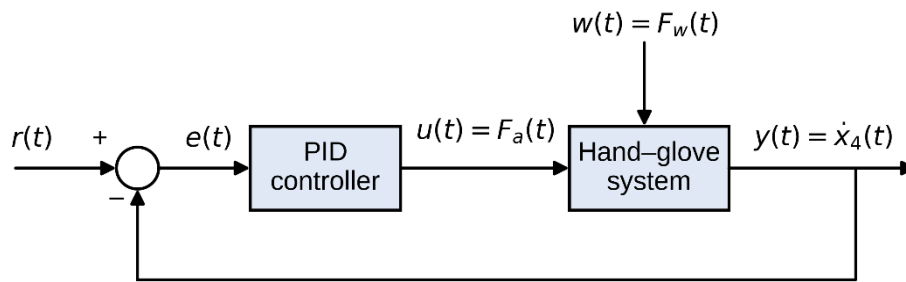


Figure 11: Feedback control of a hand–glove system with a Proportional–integral–derivative(PID) controller [Raj21].

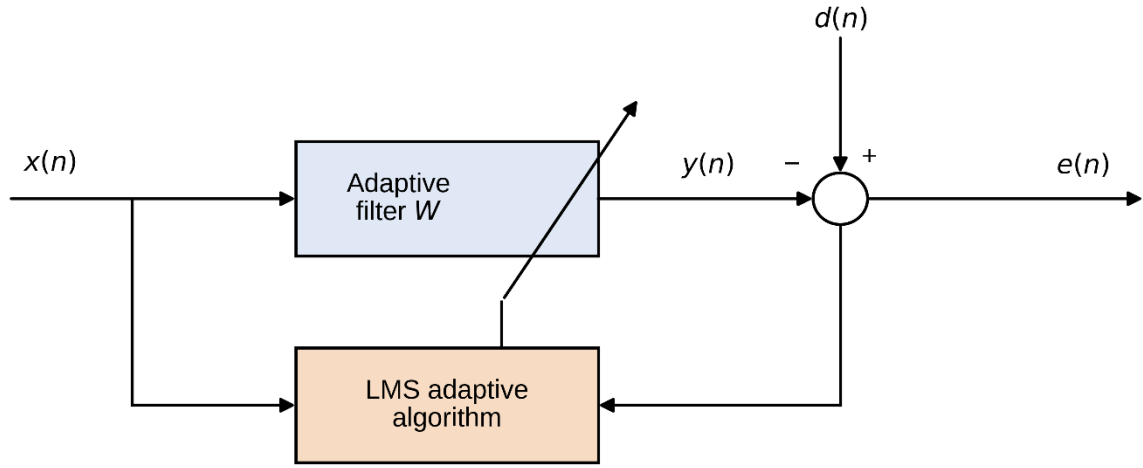
In this architecture, the measured response is returned to the controller and used to determine the subsequent control action [Raj21]. This allows the control action to respond to changes observed at the controlled location without requiring a separate measurement of the incoming disturbance [Ell87].

If a signal correlated with the disturbance is available before its effect reaches the controlled location, that information can instead be used to determine the control action in advance [Ell87].

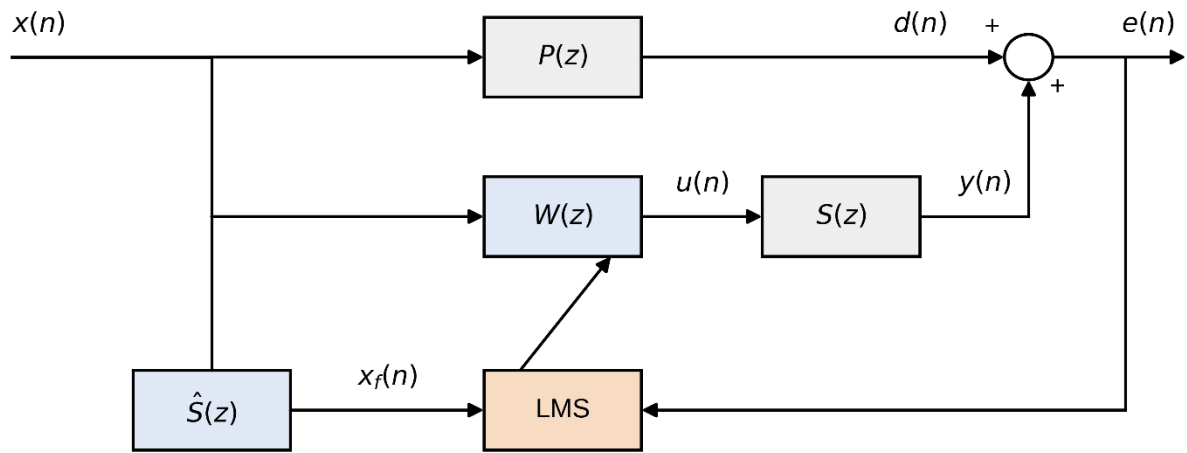
2.5.3 Feedforward and Adaptive Control

Feedforward control uses a reference signal correlated with the disturbance to determine the actuator command before the disturbance produces its full effect at the controlled location [Ell87]. Adaptive filtering extends this principle by modifying the controller during operation according to the residual error [Wid75; Ell87].

The least-mean-squares (LMS) algorithm provides the basic adaptive principle by updating filter coefficients according to the residual error [Wid75; Zhao22]. Figure 12 compares this basic LMS structure with the filtered-x least-mean-squares (FxLMS) structure used when the controller output acts through a physical secondary path.



(a) Basic LMS adaptive filter



(b) FxLMS structure for active vibration control

Figure 12: Adaptive filtering structures: (a) LMS adaptive filter [Zhao22]; (b) FxLMS control with secondary-path compensation [Yan18].

The key additional element in Figure 12(b) is the secondary-path estimate. In an active vibration-control system, the controller output passes through the amplifier, actuator and mechanical transmission path before its effect is measured; this transfer is referred to as the secondary path [Bja95; Pu19]. FxLMS filters the reference through an estimate of this path before updating the adaptive controller [Ell87; Bja95].

Narrowband FxLMS can focus this adaptation on dominant periodic vibration components, while multichannel formulations extend the same principle to several actuator–sensor paths [Liu17; Heo26]. When the secondary path changes during operation, its estimate can also be updated online [Pu19].

2.6 Vibration Measurement and Signal Processing

The reference and response signals used for vibration control must represent the underlying motion with sufficient accuracy. This depends on the selected measurand and sensor, the quality of the measurement chain and the processing used to extract the required vibration information [Alb08; ISO184311].

2.6.1 Vibration Measurands and Measurement Technologies

Mechanical vibration can be described through displacement, velocity or acceleration [ISO184311]. Acceleration is particularly suitable for compact embedded measurement and Microelectromechanical-system (MEMS) accelerometers provide a small and lightweight means of measuring it directly [Alb08]. Their performance is characterised by parameters such as measurement range, sensitivity and usable frequency response [Alb08].

Vibration can also be measured without attaching a sensor to the vibrating structure. Laser Doppler vibrometry determines surface motion optically from the Doppler shift of reflected laser light and therefore provides a non-contact alternative [Rot17]. Figure 13 compares the contact MEMS-accelerometer principle with non-contact laser Doppler vibrometry.

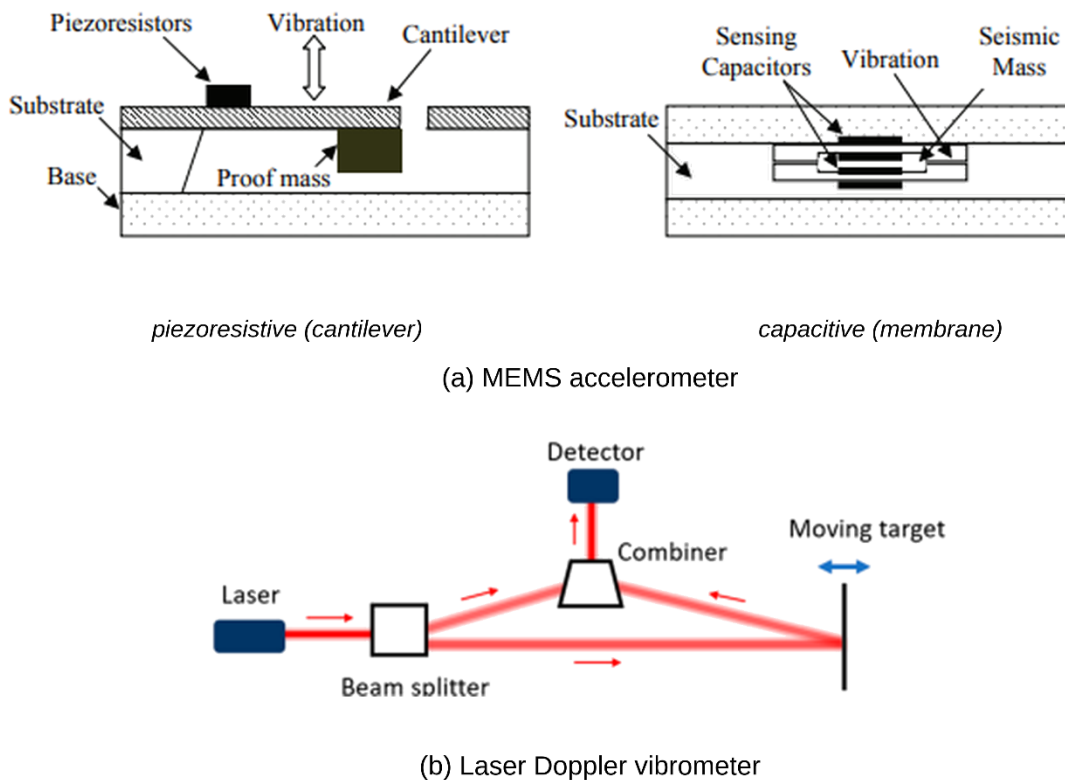


Figure 13: Vibration-measurement principles: (a) MEMS accelerometer [Alb08]; (b) laser Doppler vibrometer [Li22].

For the present measurement context, the important distinction is that an accelerometer measures motion locally through an attached inertial element, whereas laser vibrometry retrieves surface motion remotely [Alb08; Li22]. Consequently, sensor position,

orientation and mounting become part of the measurement chain for contact measurements, and measurements at different locations are not necessarily equivalent [ISO53491; ISO53492; Cle21]. Once the sensing arrangement has been defined, the reliability of the resulting signal depends on the quality of the complete measurement chain.

2.6.2 Measurement Quality, Calibration and Data Acquisition

Reliable vibration data require an appropriate sensor range and frequency response together with secure mounting and consistent orientation [Alb08; ISO5348]. Transverse sensitivity can introduce contributions from vibration outside the intended sensing direction, while sensor and acquisition noise limit the smallest vibration that can be resolved [ISO1606331; Moh03].

Calibration establishes the relationship between sensor output and a known mechanical input. Dynamic calibration can be performed by applying a controlled vibration and comparing the sensor response with a calibrated reference [ISO1606321]. This provides a means of verifying the complete measurement arrangement rather than relying only on nominal sensor specifications.

The measured signal must then be digitised at a sampling rate appropriate for the frequencies of interest, with filtering used to prevent unwanted higher-frequency components from entering the sampled data [Alb08; ISO184311]. When several channels are compared, synchronised acquisition is particularly important because timing offsets introduce relative phase errors [Che22].

Accurate sampled data can then be processed to identify the frequency content of the vibration and, where required, determine the amplitude and phase of individual components.

2.6.3 Vibration Signal Analysis and Parameter Estimation

A recorded acceleration signal first describes how vibration changes with time. Quantities such as peak and root-mean-square acceleration provide compact measures of its magnitude, but they do not reveal which frequencies contribute to the measured response [ISO184311].

Frequency-domain analysis addresses this by representing the signal according to its frequency components. The Discrete Fourier Transform (DFT) provides this frequency representation, while the Fast Fourier Transform (FFT) is an efficient algorithm for evaluating the DFT [ISO184311].

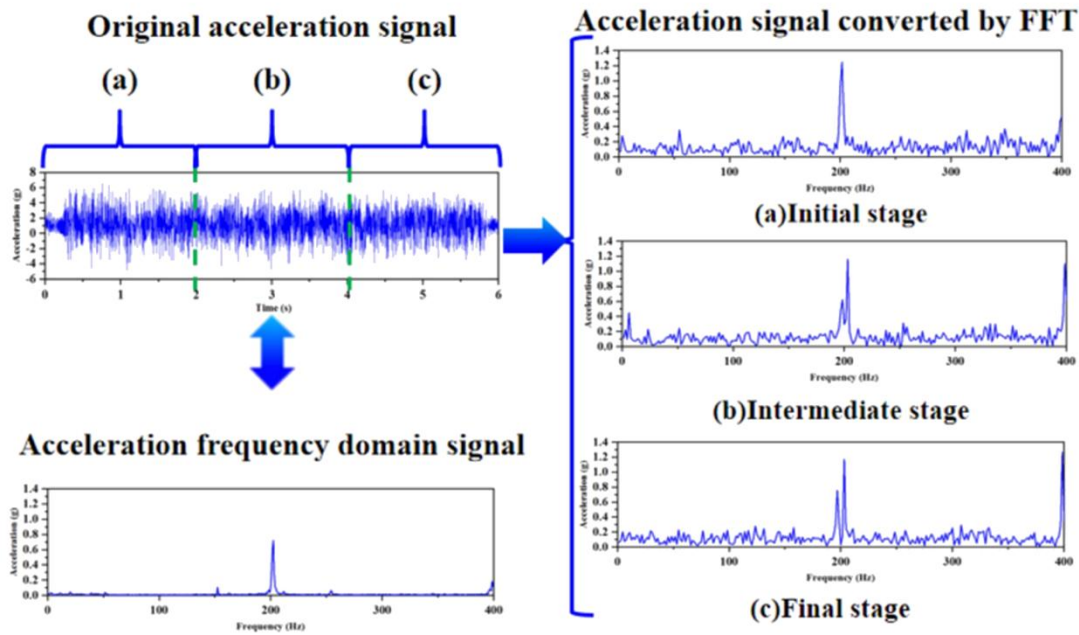


Figure 14: Measured acceleration signal and corresponding FFT spectra for (a) initial, (b) intermediate and (c) final sections [Li24].

In Figure 14, spectral peaks that are difficult to distinguish in the raw acceleration record become identifiable in the frequency domain. This is the principal reason for using frequency-domain analysis when a dominant vibration component must be located [Li24].

Finite measurement records can cause spectral leakage, so window functions are commonly applied when estimating individual frequency components [ISO184312; Har78]. Once a frequency of interest has been identified, its amplitude and phase can be estimated using quadrature detection [Ric23]. Figure 15 shows the corresponding processing structure.

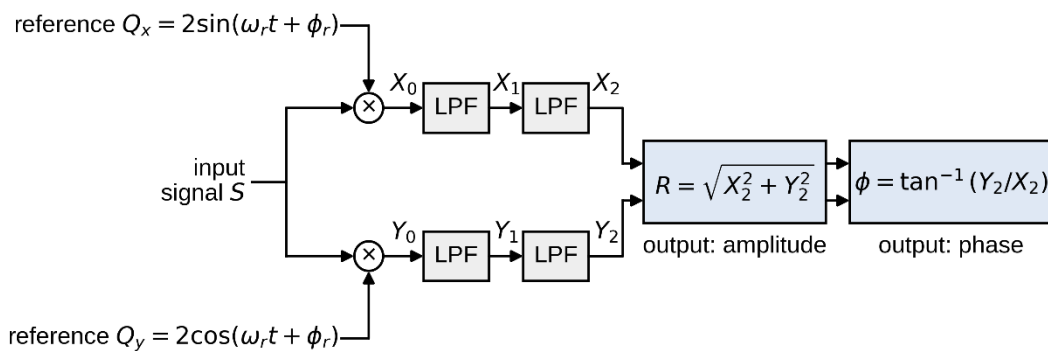


Figure 15: Quadrature estimation of amplitude and phase at a reference frequency [Har23].

The measured signal is multiplied by two orthogonal reference signals and low-pass filtered to obtain the in-phase and quadrature components. These components are then combined to determine the amplitude and relative phase at the selected reference frequency [Har23]. FFT and quadrature analysis therefore provide complementary information: spectral analysis identifies which frequency components are present, while quadrature processing determines the amplitude and phase of a selected component.

Together with reliable vibration measurement, these quantities provide the basis for characterising the behaviour of a complete vibration-control system.

2.7 Synthesis and Research Gap

Taken together, the literature shows that hand–arm vibration transmission depends on excitation frequency and direction as well as mechanical coupling and measurement location [Don21; Xu21]. Existing wearable mitigation addresses this predominantly through passive isolation or damping, whose performance remains dependent on fixed mechanical properties and prevailing coupling conditions [Don04; Ham18; Hew16]. Semiactive approaches add tunability by varying properties such as stiffness or damping, but do not independently generate the counter-vibration required for active cancellation [Spe97]. Wearable systems have also been developed that modify the mechanical interaction between the user and the vibrating tool. Ciullo et al. investigated a supernumerary robotic limb in which the tool is held by a robotic hand incorporating a damping wrist, thereby reducing the vibration transmitted to the user during drilling [Ciu21].



Figure 16: Supernumerary robotic limb with damping wrist during powered-tool operation [Ciu21].

Its mitigation principle, however, differs from the approach investigated in this thesis because the vibration is reduced primarily by damping and by changing the mechanical load path between the tool and the user rather than by generating an actively controlled counter-vibration at the hand.

Active vibration control provides a different approach by generating a mechanical response that can be adjusted according to the measured vibration. This principle has been demonstrated experimentally in active tool handles and investigated in model-based hand–glove systems [Maz17; Raj21].

However, within the literature reviewed here, experimental evidence remains limited for a compact hand-worn system that combines vibration sensing, active actuation and adaptive control to generate counteracting vibration directly at the hand during powered-tool operation. This constitutes the central research gap addressed in this thesis.

The purpose of the present work is therefore not to establish active control as inherently superior to passive protection, but to investigate the feasibility of this less extensively validated wearable architecture. Realising such a system requires sensing, actuation, control and mechanical integration to operate together within a functional prototype.

2.8 Product Development Methodologies

Systematic product-development methods provide a structured route from the identified requirements to a technically justified concept. For this work, the approaches of Pahl and Beitz and VDI 2221 are particularly relevant because they support functional decomposition, concept generation and systematic concept selection [Pah07; VDI22211].

2.8.1 Systematic Engineering Design according to Pahl and Beitz

Pahl and Beitz divide engineering design into task clarification, conceptual design, embodiment design and detailed design [Pah07]. During conceptual design, the required functions are identified and suitable working principles are combined into alternative solution concepts. Morphological analysis supports this process by organising possible solutions for each function so that complete concept variants can be generated systematically [Pah07].

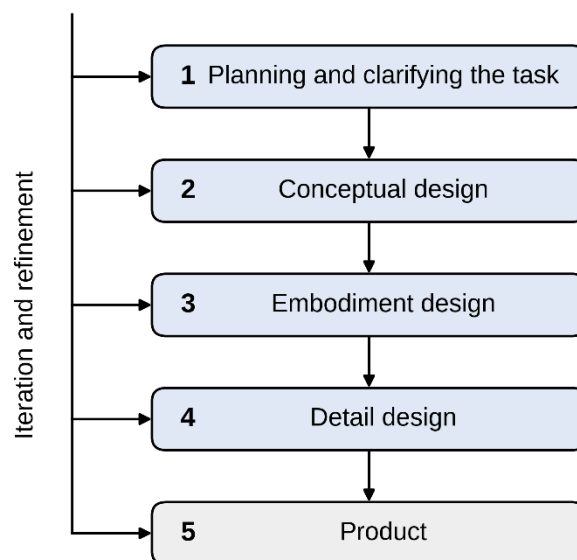


Figure 17: Main phases of systematic engineering design according to Pahl and Beitz [Pah07].

The iterative character of the process allows earlier design decisions to be revisited as the solution and its constraints become more clearly defined [Pah07].

2.8.2 VDI 2221 — Systematic Development of Technical Products and Systems

VDI 2221 provides a general framework for developing technical products from an initial task and requirements through functional analysis, solution generation, evaluation and

further development [VDI22211]. Specific design methods can be incorporated within this framework according to the needs of the development task.

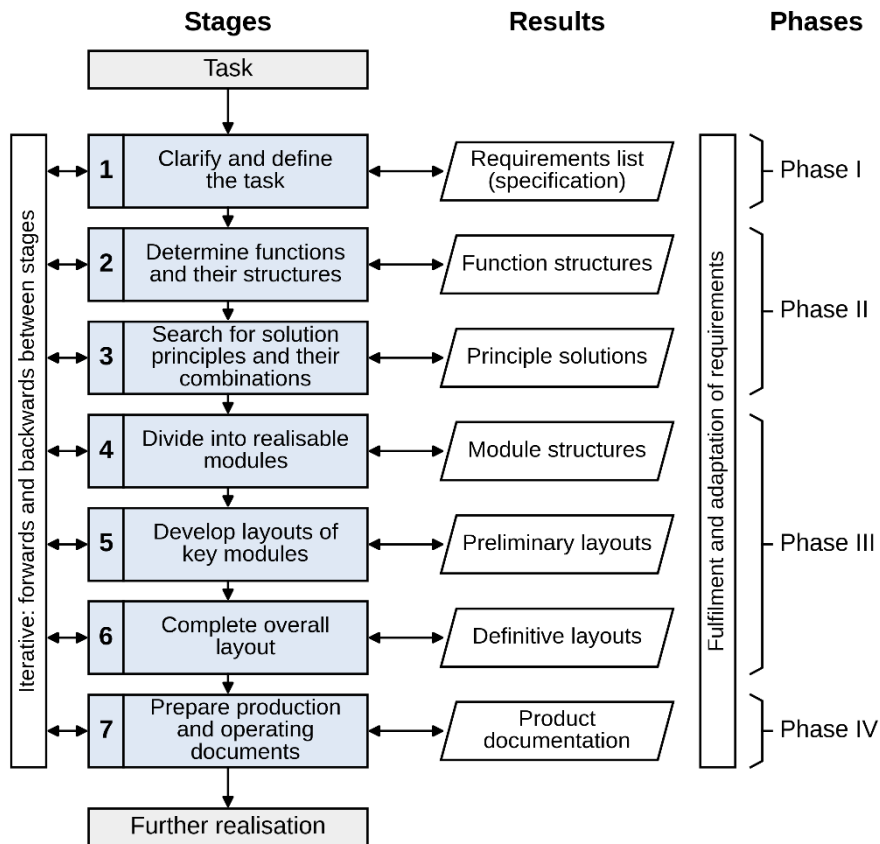


Figure 18: Development procedure according to VDI 2221 [VDI22211].

Within this framework, morphological analysis can support concept generation, while VDI 2225 Blatt 3 provides a structured basis for comparing alternatives using weighted technical and economic criteria [VDI22253]. Accordingly, VDI 2221 is used in Chapter 3 as the overall development framework, with morphological analysis supporting concept generation and weighted evaluation supporting concept selection.

3 Methodology

The technical principles identified in Chapter 2 are converted into a wearable active vibration-control prototype using VDI 2221. Requirements and functions are defined first, alternative concepts are generated and evaluated, and the selected architecture is then developed, integrated and tested [VDI22211]. Development tests used to refine the system are distinguished from the fixed-condition tests used for performance validation. A consolidated summary of the principal measurement, signal-processing, control and experimental parameters used throughout the methodology is provided in Appendix A.

3.1 Systematic Product Development

The systematic development begins by defining the main requirements of the wearable vibration-control system. These requirements are translated into the functions required to realise the overall task. Alternative technical principles are then identified for the individual functions and combined into complete concept variants before the most suitable concept is carried forward for detailed development.

3.1.1 List of Requirements

The requirements define what the wearable prototype must achieve and the conditions required for active vibration cancellation.

The primary aim is to generate an opposing vibration that reduces the dominant vibration component reaching the hand. Comparable wearable systems report reductions of about 15–60 % [Ciu21; Che23a]. For anti-vibration gloves, ISO 10819 requires a high-frequency transmissibility of at most 0.60, corresponding to an average reduction of at least 40 % over the band from 200 to 1250 Hz [ISO10819; Hew16]. The dominant vibration investigated in this work lies within this frequency range. A 40 % reduction was therefore adopted as an internal design benchmark for the controlled narrowband component. This benchmark does not constitute an ISO 10819 compliance criterion because the present metric and experimental configuration differ from the standard glove-transmissibility test.

Active cancellation requires information about both the vibration produced by the tool and the vibration remaining at the hand. The system must therefore measure the vibration at both locations. Hand–arm vibration acts in three directions [ISO53491], so all three must be measured to assess how the control affects the vibration reaching the hand. At the identified vibration frequency, the system must produce a counter-vibration whose amplitude and phase can be adjusted independently, since both determine whether cancellation or amplification occurs.

Real-time operation means that vibration measurement, processing and adjustment of the control output occur while the tool is running. The controller must therefore

continuously respond to the measured vibration without interrupting the generated counter-vibration.

The wearable structure must keep the palm and fingers free for normal gripping. All components worn on the hand must remain securely attached, because changes in their position or contact alter the mechanical path through which the counter-vibration is transmitted.

Table 1: Design requirements for the wearable active vibration-control system.

Parameter	Requirement
Vibration reduction	Reduction of the dominant vibration component at the hand by at least 40 %.
Vibration measurement	Measurement at the tool and at the hand in three directions.
Active control capability	Counter-vibration at the tool frequency with adjustable amplitude and phase
Wearability and ergonomics	Free palm and fingers for normal gripping.
Mechanical integration	Secure, repeatable attachment of all worn components
Realtime control	Continuous measurement and control update during tool operation.

Based on these requirements, the overall vibration control task was next divided into the functions required to realise the system.

3.1.2 Functional Structure

The resulting requirements were translated into the functions required to realise active vibration cancellation. At this stage, the functional structure describes what the system must achieve, without defining the technical principles used to perform these functions.

Figure 19 represents the overall task as a black box. Mechanical vibration from the handheld tool and electrical energy enter the system. The overall function is to cancel the dominant vibration component transmitted to the hand, resulting in reduced vibration at the hand while maintaining normal tool operation.

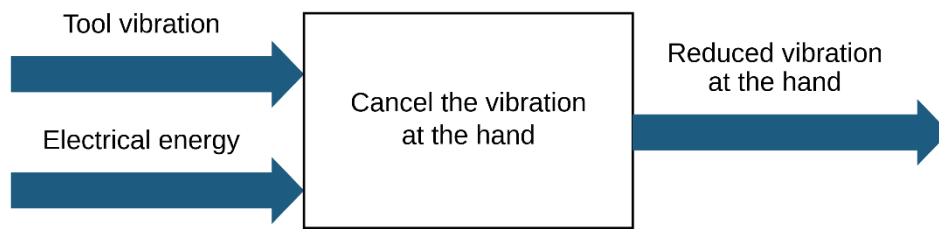


Figure 19: Black-box representation of the overall vibration-control function.

To show how this task can be realised, Figure 20 divides it into three main functions, following the order in which the system must act: gather information, decide and act. The system first acquires and processes vibration information (TF1) by measuring the vibration and determining its characteristics. From this information, it determines the required control action (TF2). Finally, it applies a mechanical action (TF3) by generating a counter-vibration and transferring it to the hand. Wearable integration and energy supply support this functional chain throughout operation.

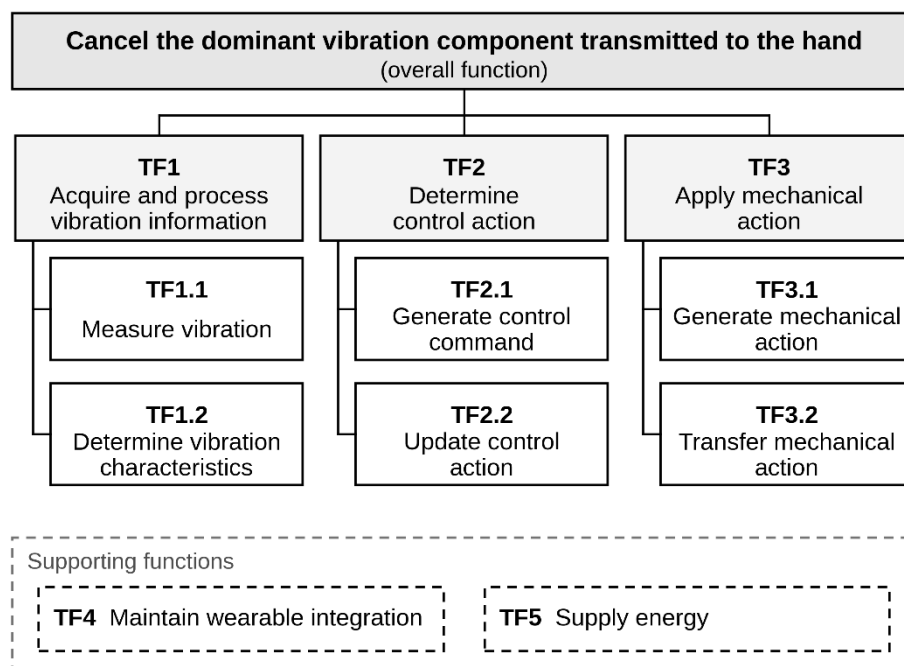


Figure 20: Functional structure of the wearable active vibration-control system.

The functional structure therefore defines the tasks that must be fulfilled independently of a particular technical solution. This provides the basis for comparing alternative working principles.

3.1.3 Morphological Analysis and Concept Variants

Alternative technical principles were identified for each subfunction and organised in a morphological box [Pah07]. The alternatives were defined at the level of working principles rather than individual commercial components.

Table 2 shows the considered options for vibration measurement and processing, control generation and update, mechanical actuation and transfer, wearable integration and energy supply. In this way, each function defined in Figure 20 is linked to several possible technical solutions.

Table 2: Morphological box of solution principles for the system subfunctions.

Subfunction	A	B	C	D
Measure vibration	Acceleration measurement	Contact velocity measurement	Relative displacement measurement	Noncontact optical measurement
Determine vibration characteristics	Time domain estimation	FFT/DFT analysis	Phase locked tracking	FFT/DFT quadrature estimation
Generate control command	Prescribed harmonic control	Feedback Control	Feedforward control	Combined feedforward–feedback
Update control action	Rule/gain scheduled update	Search/optimisation based update	Model/estimator based update	Error driven adaptive update
Generate mechanical action	LRA	Electrodynamic inertial	Piezoelectric actuator	
Transfer mechanical action	Direct rigid coupling	Preloaded stiff coupling	Frame mediated transfer	Mechanically amplified transfer
Maintain wearable integration	Hand/glove mounting	Wrist mounting	Forearm mounting	Hand–wrist/forearm support
Supply energy	External electrical supply	Rechargeable battery	Replaceable battery	

Not every combination in Table 2 forms a workable system. The sensing, control, actuation and mechanical-integration principles must operate together as one architecture. Compatible principles were therefore combined into five complete concept variants, as summarised in Table 3.

Table 3: Concept variants and their solution principles.

Variant	Measurement & processing	Control	Actuation & coupling	Wearable integration	Energy
V1 Resonant feedback	Acceleration + phase locked tracking	Feedback + gain scheduling	LRA + direct rigid coupling	Wrist	Rechargeable battery
V2 Adaptive electrodynamic	Acceleration + FFT/DFT + quadrature	Feedforward + adaptive update	Electrodynamic inertial + direct coupling	Hand/wrist	External supply
V3 Piezoelectric feedback	Acceleration + time domain estimation	Feedback + model based update	Piezoelectric + preloaded stiff coupling	Forearm	External supply
V4 Harmonic electrodynamic	Acceleration + FFT/DFT	Prescribed harmonic + optimisation	Electrodynamic inertial + direct coupling	Wrist	External supply
V5 Hybrid electrodynamic	Acceleration + phase locked tracking	Feedforward feedback + model based update	Electrodynamic inertial + frame mediated transfer	Hand–wrist/forearm	External supply

The five variants represent different ways of realising the same functional chain. They range from resonant and prescribed-harmonic concepts to adaptive and hybrid control architectures, while also differing in actuation, mechanical coupling and wearable integration. The variants are therefore compared as complete system concepts, rather than as isolated component choices.

The complete concepts were then evaluated against the original requirements to determine which architecture should be carried forward for detailed development.

3.1.4 Concept Evaluation and Decision Matrix

The five concept variants were evaluated in two steps. First, each variant was checked against the requirements in Table 1. All five could in principle fulfil the requirements and

were therefore retained for comparative evaluation. Second, the variants were compared using weighted evaluation criteria according to VDI 2225 [VDI22253].

The evaluation criteria were derived from the requirements, so that each criterion measures how well a concept can fulfil one or more of them. Table 4 shows this derivation. The weight g reflects the importance of each criterion. Criteria that decide whether the main function can be achieved at all receive the highest weight of 5. Criteria directly linked to supporting requirements receive a weight of 4. Criteria that determine how reliably and feasibly a concept can be realised receive a weight of 3, while cost, treated as a secondary practical constraint, receives a weight of 2. The requirement for vibration measurement does not appear as a criterion, because all five variants use the same principle, acceleration measurement, and it therefore does not distinguish between them.

Table 4: Evaluation criteria and weighting for concept selection.

No.	Criterion	Derived from	g	Reason for weight
C1	Vibration control	Vibration cancellation	5	Main function of the system
C2	Adaptability	Active control; real-time operation	4	Counter-vibration must follow changes during operation
C3	Actuator capability	Vibration cancellation; active control	5	Sufficient counter-vibration is a precondition for any cancellation
C4	Mechanical integration	Mechanical integration	4	Defined transfer of the counter-vibration to the hand
C5	Wearability	Wearability	4	Tool must remain usable
C6	Implementation effort	Practical realisation	3	Feasibility within the available hardware and development scope
C7	Robustness	Active control; mechanical integration	3	Consistent operation despite changes in the mechanical connection
C8	Cost	Practical constraint	2	Not a requirement, but relevant for realisation

Each variant was rated using a performance score p_{ij} from 0 to 4, where 0 represents insufficient fulfilment and 4 represents very good fulfilment. For implementation effort and cost, a higher rating represents lower implementation effort and lower expected cost, respectively. The overall score S_j of each variant was calculated as

$$S_j = \frac{\sum_{i=1}^n g_i p_{ij}}{\sum_{i=1}^n g_i} \quad (1)$$

where g_i is the importance factor of criterion i and p_{ij} is the rating of concept j for that criterion. Normalisation by the sum of the importance factors retains the final score on the 0–4 rating scale. Table 5 shows the resulting decision matrix.

Table 5: Weighted decision matrix for the concept variants.

No.	Evaluation Criterion	g	V1		V2		V3		V4		V5	
			p	g·p	p	g·p	p	g·p	p	g·p	p	g·p
C1	Vibration control	5	3	15	4	20	2	10	3	15	4	20
C2	Adaptability	4	2	8	4	16	2	8	2	8	4	16
C3	Actuator capability	5	2	10	4	20	1	5	4	20	4	20
C4	Mechanical integration	4	3	12	3	12	2	8	3	12	2	8
C5	Wearability	4	4	16	3	12	2	8	3	12	1	4
C6	Implementation effort	3	4	12	2	6	2	6	4	12	1	3
C7	Robustness	3	3	9	3	9	2	6	3	9	2	6
C8	Cost	2	4	8	3	6	2	4	3	6	1	2
$\Sigma g \cdot p$		30	90		101		55		94		79	
Overall score $S = \Sigma g \cdot p / 30$			3.00		3.37		1.83		3.13		2.63	
Rank			3		1		5		2		4	

The ratings represent the expected suitability of each complete concept rather than measured performance. Table 5 shows that V2 achieves the highest weighted score of 3.37, followed by V4 (3.13), V1 (3.00), V5 (2.63) and V3 (1.83). V2 receives the maximum rating in the three control-related criteria C1–C3, combining high expected vibration-control potential, adaptability and actuator capability. V4 offers comparable actuator capability with lower implementation effort, but its prescribed harmonic control is less adaptable. V1 is simpler and more wearable but is limited by its resonant actuation principle, while V5 introduces greater mechanical and control complexity. V3 is limited mainly by the available mechanical displacement of the piezoelectric actuator.

Variant V2 was therefore carried forward for detailed development. It combines acceleration-based vibration measurement, adaptive feedforward control and electrodynamic actuation.

3.2 Design and Development

Active cancellation at the hand requires a counter-vibration that matches the dominant tool vibration in frequency and amplitude and opposes it in phase. The main challenge is that this counter-vibration reaches the hand through a mechanical connection whose response can change with the wearable attachment and hand–tool interaction.

Concept V2 was therefore developed by addressing a sequence of engineering questions: where the vibration should be measured and the counter-vibration applied, which vibration component should be targeted, whether the actuator can reproduce it, how the mechanical transmission to the hand affects the response, and how the control action can adapt accordingly. Development therefore proceeds from system architecture

and measurement verification to control-frequency identification, actuator characterisation, amplitude and phase investigation, secondary-path identification and adaptive control.

3.2.1 System Architecture and Wearable Integration

Active cancellation requires three functions to act together: the vibration produced by the tool must be measured, an opposing mechanical vibration must be generated, and the vibration reaching the hand must be observed. The first design task was therefore to define where these functions act and how the sensing, control and actuation elements are connected. Figure 21 shows the resulting architecture and the implemented components.

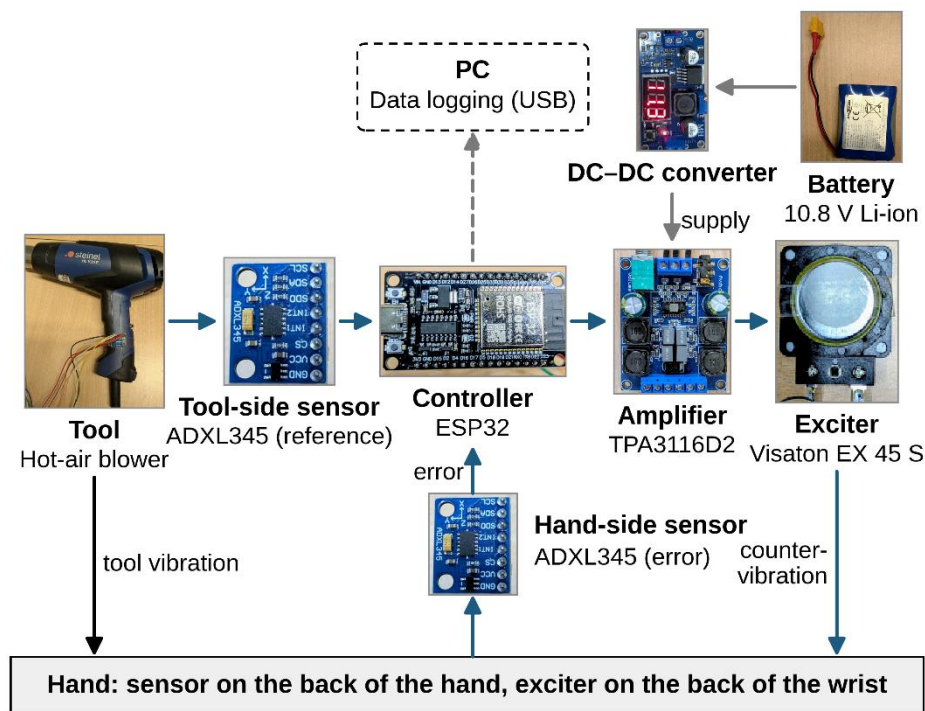


Figure 21: System architecture and signal flow of the wearable active vibration-control prototype.

A tool-side ADXL345 mounted on the tool grip measures the incoming vibration and provides the reference signal. A second ADXL345 on the back of the hand measures the vibration at the control location and provides the error signal. The ESP32 processes these measurements and generates the actuator command. This command is amplified by the TPA3116D2 and applied to the Visaton EX 45 S exciter, which introduces the mechanical counter-vibration at the back of the wrist. The amplifier is supplied from a 10.8 V lithium-ion battery through a Direct-current-to-direct-current converter. The personal computer (PC) is used for test supervision and data logging and is not part of the real-time control loop.

Because the actuator output reaches the hand through the amplifier, exciter, attachment and wrist–hand coupling, the response measured at the hand differs in amplitude and phase from the electrical command. This actuator-to-hand transmission is the secondary path considered later in the control development.

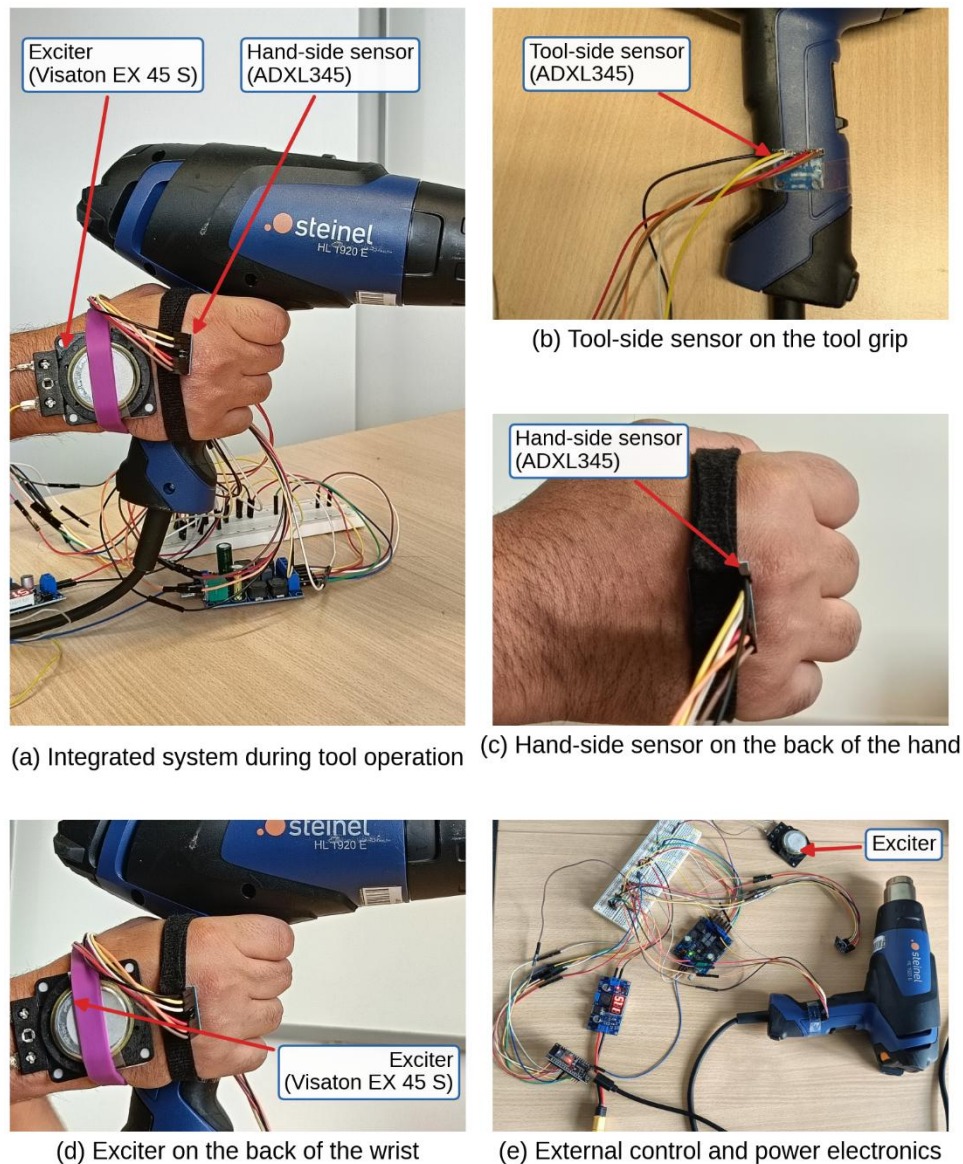


Figure 22: Wearable active vibration-control prototype: (a) integrated system during tool operation, (b) tool-side sensor, (c) hand-side sensor, (d) wrist-mounted exciter and (e) external control and power electronics.

The tool-side sensor is mounted on the tool grip (Figure 22b), the hand-side sensor is secured to the back of the hand using an adjustable strap (Figure 22c), and the exciter is held against the back of the wrist by an elastic attachment (Figure 22d). The palm and fingers remain free for normal gripping (Figure 22a), satisfying the requirement that these areas remain unobstructed during tool operation. The controller, amplifier and supporting power electronics remain externally mounted on the test bench and are connected to the wearable elements by cables (Figure 22e).

The resulting architecture therefore fixes the three physical roles used throughout the development: the tool-side sensor measures the disturbance, the hand-side sensor measures the vibration to be reduced, and the wrist-mounted exciter provides the counter-vibration. Reliable control therefore depends on consistent vibration measurements at both the tool and the hand.

3.2.2 Vibration Measurement and Embedded Hardware

A compact embedded measurement system was therefore required to measure both the vibration produced by the tool and the vibration reaching the hand consistently.

Measurement hardware: Two triaxial ADXL345 accelerometers were used with identical measurement settings. The ± 16 g full-resolution range was chosen to provide sufficient headroom for acceleration peaks without clipping. An output data rate of 1600 Hz was used to resolve the dynamic vibration encountered during the development tests. Both accelerometers were connected to the ESP32 through a shared Serial Peripheral Interface (SPI) bus with separate chip-select lines, allowing repeated acquisition from both triaxial sensors during operation. The same ESP32 was then used for the subsequent signal-processing and control functions.

Before dynamic testing, both accelerometers were checked in six static orientations by aligning each positive and negative axis with gravity. Three repeated measurements were recorded in every orientation. Figure 23(a) shows that both sensors produced the expected positive and negative response for the corresponding axis, while the repeated measurements showed consistent behaviour. The static check therefore confirmed repeatable sensor response and correct axis orientation for the subsequent development tests.

The next check examined whether both measurement channels identified the same vibration when exposed to a common dynamic input. The two accelerometers were temporarily mounted at different positions on the same excited structure and subjected to a controlled 268 Hz sinusoidal excitation over 12 independent cycles. The purpose of the test was straightforward: when both sensors experience the same periodic excitation, both measurement chains should identify the same frequency.

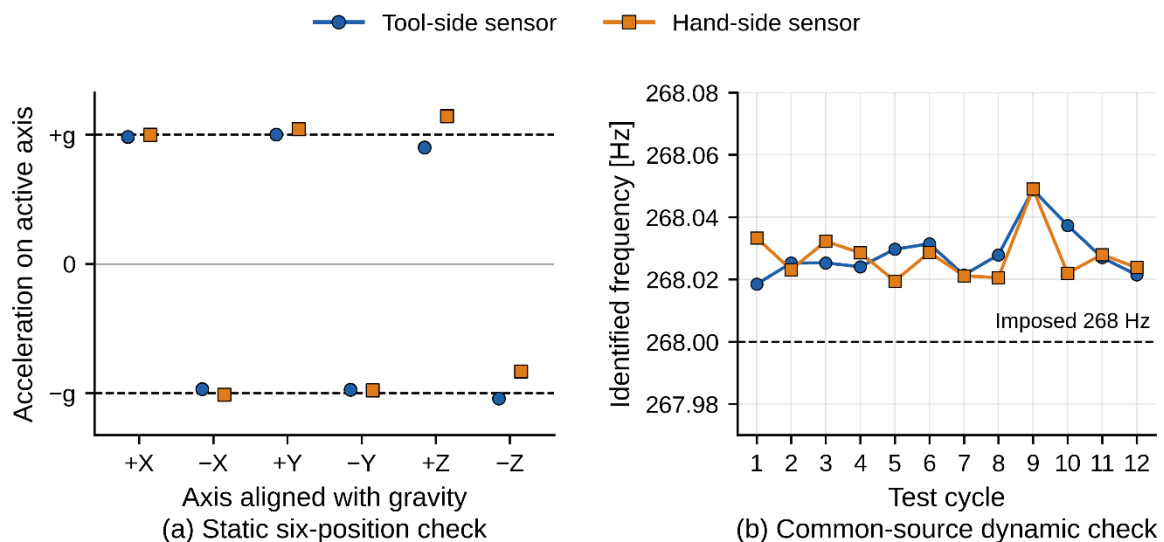


Figure 23: Dual-accelerometer verification: (a) static gravity check; (b) common-source dynamic frequency check.

Figure 23(b) shows that the frequencies identified by the two channels were almost identical in every test. The mean measured frequencies were 268.028 Hz and 268.027 Hz, with a mean absolute run-wise difference of approximately 0.006 Hz.

The dual-accelerometer architecture was therefore retained. The tool-side sensor was used to describe the vibration entering from the tool, while the hand-side sensor measured the vibration at the location where the effect of the active system would be evaluated.

With the measurement chain verified for axis orientation and common-source frequency consistency, the tool vibration could then be screened for a repeatable component suitable for active cancellation.

3.2.3 Frequency Identification and Narrowband Estimation

The verified measurement chain was used to determine whether the tool produced a repeatable vibration component suitable for active cancellation.

The tool was first operated with the actuator switched off, and its vibration was screened over a broad frequency range in eight independent measurements so that the following development would not depend on a single run. Figure 24(a) shows several recurring frequency regions in the tool signal, but their response at the hand differed considerably. Three regions occurred in all eight measurements, and the 180–205 Hz region was even slightly stronger at the tool than the 268–272 Hz region. At the hand, however, the 268–272 Hz region produced by far the largest response, with a mean amplitude of 1.35 m/s^2 , compared with 0.41 m/s^2 or less for the other regions. Other regions were also observed repeatedly at the tool, but their hand-side response was smaller, or their occurrence was less consistent. The 268–272 Hz region was therefore chosen as the control target because it occurred in all eight tool measurements and produced the largest mean hand-side response.

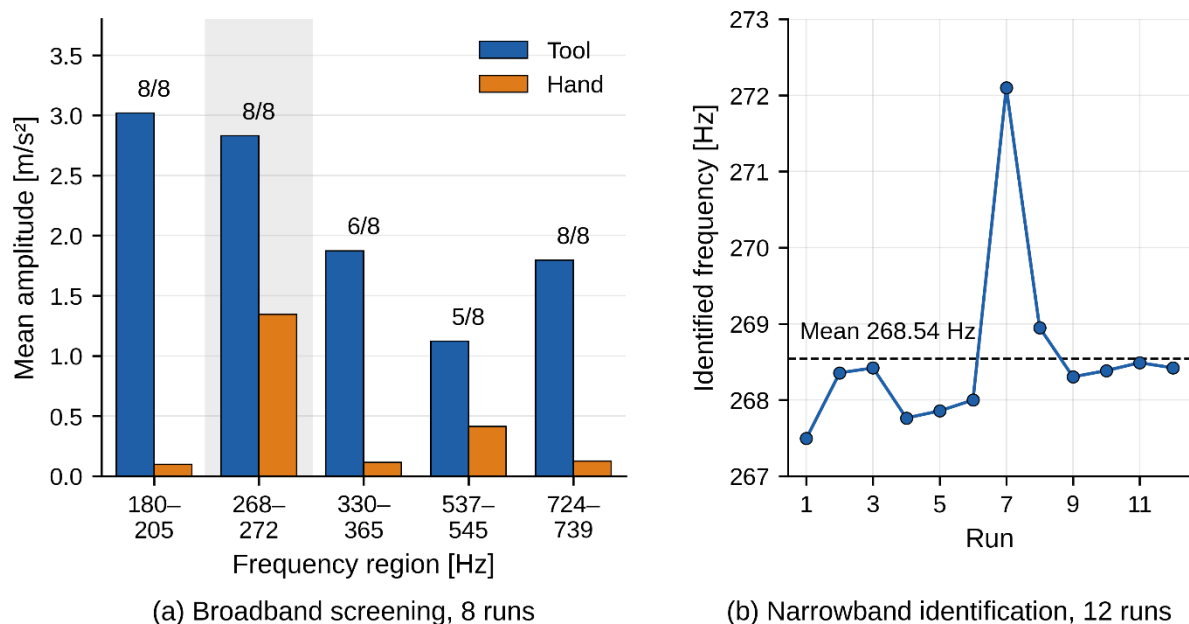


Figure 24: Selection of the control-frequency region: (a) recurring spectral regions from eight tool runs with corresponding mean tool- and hand-side amplitudes; (b) narrowband frequency estimates from 12 independent runs.

The subsequent analysis was restricted to a 240–320 Hz search window around this region. This retained sufficient margin around the observed vibration family while excluding the other recurring regions from the frequency search.

Within this window, the vibration frequency was determined in two stages. First, FFT was applied to the measured acceleration. The FFT represents the measured vibration as discrete frequency components, allowing the strongest component within the 240–320 Hz window to be located. Three 1024-sample FFT records were averaged to reduce fluctuations between individual measurements. At the 1600 Hz acquisition rate, each FFT has a frequency-bin spacing of approximately 1.56 Hz. Because the actual vibration frequency can lie between adjacent FFT bins, interpolation between neighbouring spectral values was used to refine the initial estimate.

The FFT estimate was then refined using quadrature processing. The measured vibration was multiplied by sine and cosine reference signals at the estimated frequency, providing the amplitude and phase of that component. A 2560-sample record was divided into ten blocks of 256 samples, each corresponding to approximately 160 ms. If the reference frequency differed slightly from the actual vibration frequency, the calculated phase changed progressively from one block to the next. This phase progression was used to correct the frequency estimate. A frequency was accepted only when it satisfied the defined amplitude and phase-consistency criteria across the ten blocks. The complete identification procedure was evaluated over 12 independent tool runs. Figure 24(b) shows that a valid frequency was obtained in every run. The identified values ranged from 267.5 to 272.1 Hz, with a mean of 268.54 Hz. Eleven of the twelve runs lay between 267.5 and 269.0 Hz, while one run at 272.1 Hz showed that the exact vibration frequency could shift within the same narrow region.

Design decision: The repeated measurements showed that the vibration to be addressed occurred within a repeatable narrow frequency region, while its exact frequency varied between runs. The control procedure was therefore configured to identify the current frequency within the 240–320 Hz search window before generating the counter-vibration.

The identified vibration region consequently defined the frequency range that the actuator had to reproduce.

3.2.4 Actuation and Mechanical Action Development

The identified 240–320 Hz vibration region defined the frequency range in which the actuator had to generate a controllable mechanical response.

The counter-vibration was generated using a Visaton EX 45 S electrodynamic exciter. Because the ESP32 cannot supply the power required by the exciter directly, the command signal was passed through a TPA3116D2 power amplifier before reaching the actuator. This arrangement allowed the controller to prescribe the frequency, amplitude and phase of the mechanical excitation independently.

A phase-continuous numerically controlled oscillator (NCO) was implemented on the ESP32 to generate the sinusoidal actuator command. The NCO advances the phase of the sinusoid continuously according to the commanded frequency, allowing its frequency,

amplitude and phase to be adjusted without restarting the excitation. The command waveform was updated at 10 kHz to provide a smooth actuator input. The actuator response was then examined directly within the identified vibration region.

The exciter was operated at 268, 270 and 275 Hz, covering the region in which the tool vibration had been repeatedly identified. At each frequency, the actuator command was increased over a range of command amplitudes while the resulting acceleration was measured by the triaxial hand-side accelerometer. The same narrowband estimation procedure was used to determine the mechanical response at the commanded frequency.

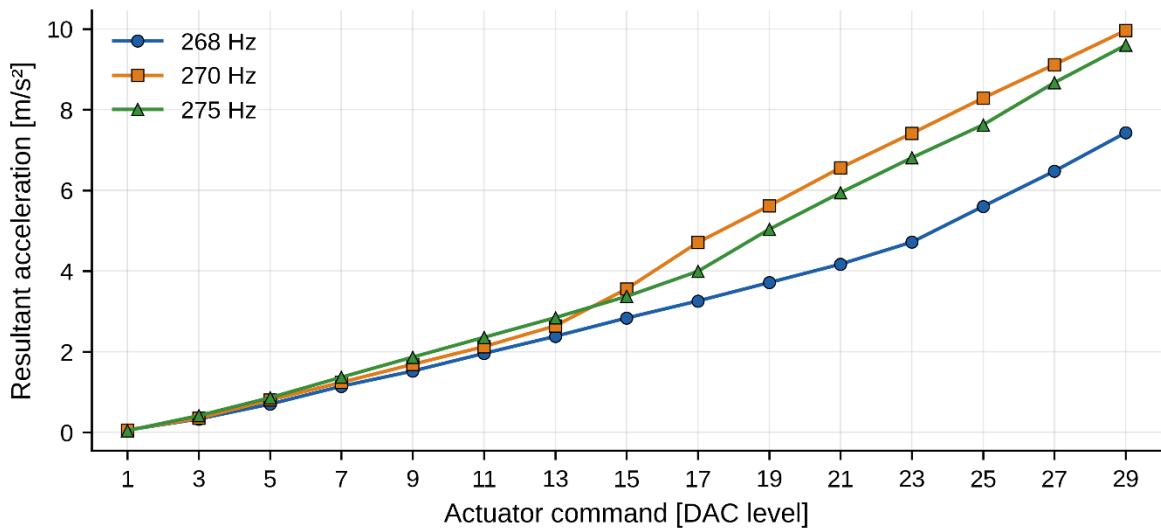


Figure 25: Hand-side resultant acceleration generated by the Visaton EX 45 S as a function of actuator command at 268, 270 and 275 Hz.

Figure 25 shows that the exciter generated a clear mechanical response at all three investigated frequencies. In each case, the measured resultant acceleration increased with actuator command, confirming that the actuator could provide adjustable mechanical excitation throughout the frequency region required for active cancellation. The exact response differed between frequencies, indicating that the vibration generated at the hand cannot be determined from the electrical command alone and must instead be measured after transmission through the actuator attachment and wearable structure.

Design decision: The Visaton EX 45 S was retained because it could generate an adjustable mechanical response within the required frequency region. The actuator command could therefore be used as the basis for matching the magnitude of the counter-vibration to the vibration measured at the hand.

3.2.5 Prescribed Harmonic Amplitude Matching

After confirming adequate actuator response within the required frequency region, its vibration magnitude had to be matched to the tool-induced vibration already reaching the hand.

Matching the magnitude is necessary before investigating phase cancellation. If the actuator-generated vibration is substantially weaker or stronger than the tool-induced

vibration, changing the phase alone cannot produce a meaningful opposing response. The actuator command therefore had to be adjusted until both vibration components had approximately the same amplitude at the hand.

The required amplitude was determined directly from the hand-side measurement. The tool was first operated with the actuator switched off, and the vibration amplitude at the identified vibration frequency was measured. This value was used as the target for the actuator. The tool was then switched off and the exciter was operated alone at the same frequency, so that its contribution could be measured independently.

A fixed conversion between digital-to-analogue converter (DAC) command and hand-side acceleration was not used because the actuator-response tests showed that the mechanical response depends on the attachment and wearing condition. Instead, the actuator command was determined from the measured acceleration. The command was increased until responses below and above the target amplitude had been obtained. These two points formed a bracket around the required value, and interpolation between them was used to estimate the next DAC command. The procedure continued until the measured actuator amplitude entered the prescribed $\pm 5\%$ acceptance range around the target.

Once the target had been reached, the matched amplitude was held and measured repeatedly to verify that it remained stable before the subsequent phase investigation.

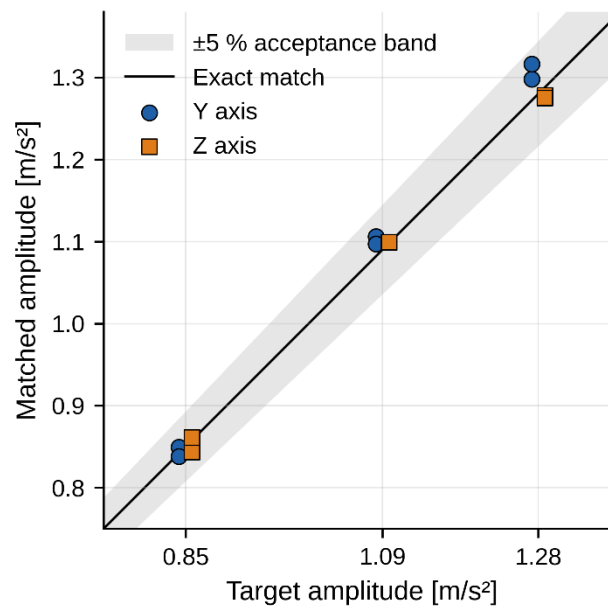


Figure 26: Amplitude-matching performance at 270 Hz with $\pm 5\%$ acceptance limits.

The matcher was verified at 270 Hz using prescribed target amplitudes of 0.85, 1.09 and 1.28 m/s² in two measurement directions. Figure 26 shows that the matched amplitudes remained close to their respective targets in all 11 verification measurements and within the $\pm 5\%$ acceptance band. This confirmed that the actuator vibration could be set reliably to a required magnitude before its phase relationship was investigated. The Y and Z direction data in Figure 26 are used only to verify the amplitude-matching routine; they do not define the control directions used later in the FxLMS experiments.

No permanent control axis was selected at this stage. The purpose of the procedure was only to establish that, for the hand-side direction being examined, the actuator magnitude could be matched reliably to the measured tool-induced vibration.

Design decision: The actuator magnitude was therefore set from the measured vibration rather than from a fixed command-to-acceleration calibration. Once the required amplitude had been established and verified, it was maintained so that the influence of the relative phase could be examined independently.

3.2.6 Open Loop Phase Cancellation and Mechanical Coupling Limitation

Once the vibration frequency and actuator magnitude were established, the remaining variable governing cancellation was the relative phase at the hand.

For two sinusoidal vibrations of the same frequency and similar amplitude, cancellation depends on their relative phase. In an ideal case, an opposing response would be obtained when the two vibrations arrive approximately 180° apart. In the present system, however, the phase commanded by the ESP32 is not necessarily the phase that reaches the hand. The actuator signal passes through the amplifier, exciter, attachment and the mechanical connection to the hand, each of which can alter the transmitted response.

The phase relationship was therefore examined with the tool and actuator operating simultaneously at the same identified vibration frequency. The actuator magnitude was first established using the amplitude-matching procedure.

The actuator phase was then varied through a complete cycle from 0° to 330° in 30° increments, while the resulting narrowband vibration was measured at the hand. A separate actuator-off measurement provided the reference vibration level against which the phase-dependent response was compared.

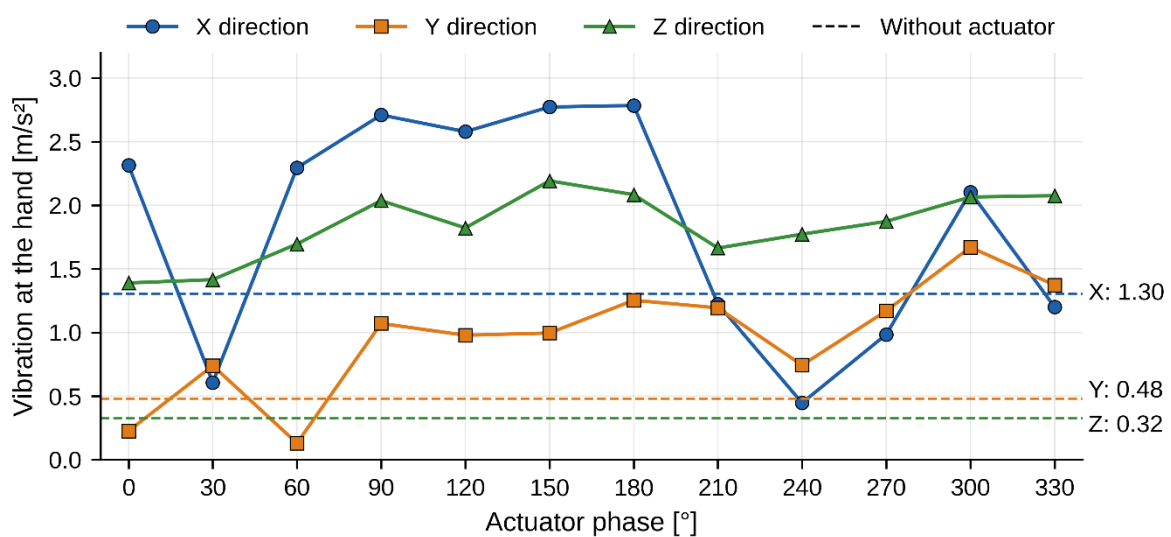


Figure 27: Hand-side narrowband vibration amplitude in the X, Y and Z directions during a 0° – 330° actuator phase sweep; dashed lines indicate the actuator-off reference amplitudes.

Figure 27 shows that changing the actuator phase could either reduce or increase the measured vibration, and that the phase associated with the lowest response was not the same for all measured directions. The required cancellation phase could therefore not be determined from the electrical command alone.

The important implication was that the actuator-generated vibration undergoes an additional magnitude and phase change before reaching the hand. This behaviour is determined by the mechanical transmission between the actuator command and the hand-side measurement and can change with the attachment and coupling condition.

Design decision: A fixed open-loop phase was therefore not retained as the final control approach. The effect of the actuator-to-hand transmission had to be included explicitly so that the controller could account for the phase and magnitude changes introduced by the mechanical system.

3.2.7 Secondary Path Identification

The phase investigation showed that the commanded vibration was altered in both magnitude and phase before reaching the hand, requiring the actuator-to-hand transmission to be identified explicitly.

This transmission is referred to as the secondary path. It includes the amplifier, exciter, attachment and the mechanical connection to the hand. Its effect can be described using two quantities. The gain tells how large the vibration measured at the hand becomes for a given actuator command, while the phase shift tells how far the vibration arriving at the hand is shifted relative to the commanded vibration.

In simple terms, the secondary-path measurement answers two questions: how much of the commanded vibration reaches the hand, and at what phase does it arrive?

To determine this, the tool was switched off and the actuator was driven at the same frequency as the identified tool vibration. A known sinusoidal command was applied and the resulting vibration was measured at the hand. The commanded and measured signals were then compared using the same quadrature amplitude and phase estimates used for frequency identification.

The amplitude comparison provided the secondary-path gain, while the phase difference provided the secondary-path phase shift. Repeated measurements were used to confirm that the identified relationship remained stable.

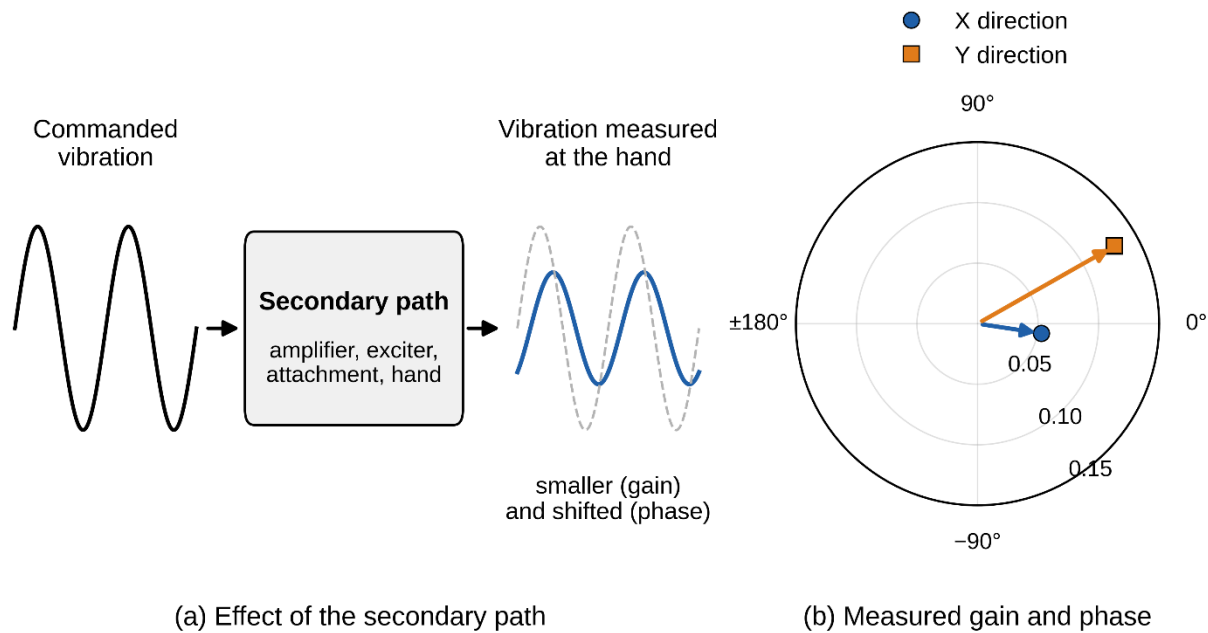


Figure 28: Secondary-path identification: (a) effect on vibration amplitude and phase; (b) measured gain and phase in the X and Y directions.

Figure 28(a) illustrates how the secondary path changes the commanded vibration before it reaches the hand. In Figure 28(b), radial position represents the secondary-path gain, while angular position represents the corresponding phase shift in the X and Y directions. The angle represents the secondary-path phase shift, showing how far the vibration arriving at the hand is shifted relative to the commanded signal.

The secondary-path identification therefore provided the controller with a direct description of how the actuator command appears at the hand. This is the information that the open-loop phase approach was missing.

Design decision: The measured secondary-path gain and phase were included in the control procedure so that the counter-vibration could be adjusted according to the actual mechanical response at the hand rather than the electrical command alone.

3.2.8 Adaptive Narrowband FxLMS Control Development

The open-loop tests showed that a fixed actuator amplitude and phase could not provide a reliable final solution because the actuator-to-hand path alters the commanded response.

A narrowband (FxLMS) controller was used for this purpose. The controller uses the tool-side vibration as a reference for the disturbance that must be opposed and the hand-side vibration as the error that must be reduced. Unlike the prescribed amplitude and phase used in the preceding steps, FxLMS continuously adjusts the actuator command from these measured signals.

In the implemented narrowband controller, the adaptive filter is represented by a single complex weight W . The magnitude of W determines the actuator-command amplitude,

while its angle determines the actuator phase. Updating W therefore adjusts amplitude and phase simultaneously.

The identified secondary-path gain and phase were included because a change commanded at the actuator does not reach the hand unchanged. The tool-side reference is therefore corrected mathematically using the identified secondary-path gain and phase before the adaptive weight is updated. This is the filtered-x principle: the controller considers how its own actuator command is expected to appear at the hand before deciding how the counter-vibration should be changed.

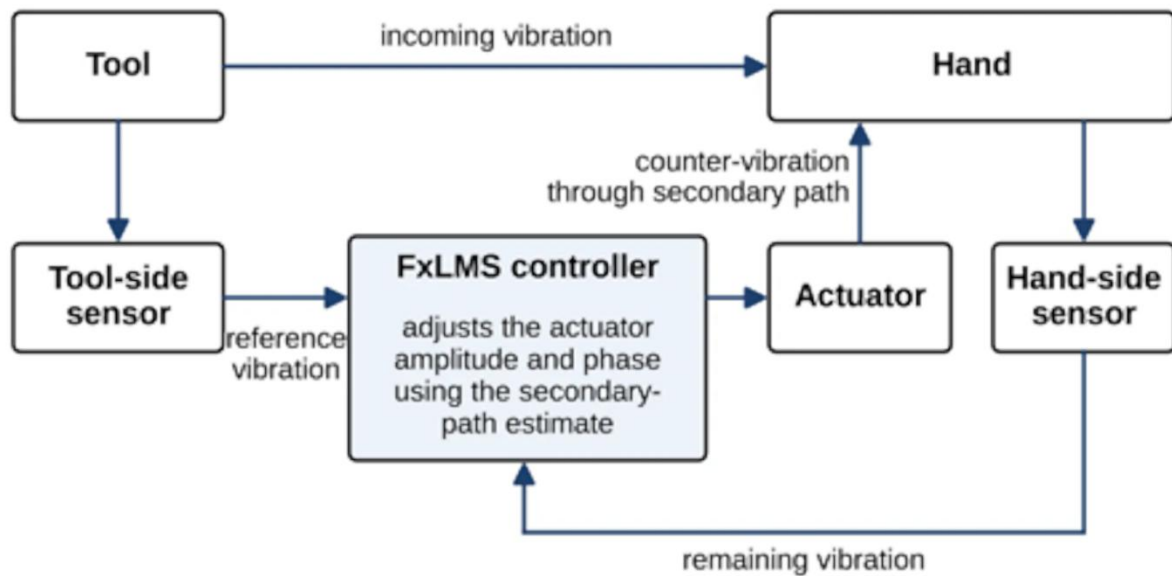


Figure 29: Operating principle of the narrowband FxLMS controller.

The tool-side sensor supplies the vibration reference, while the hand-side sensor measures the residual vibration after the actuator response has passed through the secondary path. The controller uses both measurements and the identified secondary-path information to adjust the actuator amplitude and phase.

The controller updated the counter-vibration repeatedly while the tool was operating. Every 160 ms, the current tool-side vibration and the vibration remaining at the hand were evaluated. The secondary-path estimate was applied to the reference, and the adaptive weight was then corrected according to the measured hand response. The updated weight determined the amplitude and phase supplied to the phase-continuous NCO for the next control interval. In this way, the counter-vibration could progressively adapt rather than relying on one fixed actuator setting.

Because the prototype contains only one actuator, the adaptive controller operates on one selected hand-side direction at a time. The remaining acceleration directions are still measured so that the effect of the control action outside the selected direction can later be evaluated.

Design decision: Narrowband FxLMS was retained as the final control principle because it allowed the actuator amplitude and phase to adapt from the measured vibration while

explicitly accounting for the secondary path between the actuator and the hand. The controller therefore no longer required a prescribed open-loop amplitude or phase.

With the adaptive control principle established, the final engineering question was how frequency identification, secondary-path information, adaptive control and the required operating checks should be combined into one complete control sequence.

3.2.9 Integrated Control Logic and Operating Sequence

The validated sensing, frequency-identification, secondary-path and adaptive-control functions were then combined into one complete operating sequence.

The operating sequence begins with hardware initialisation, during which the two accelerometers and the phase-continuous actuator signal generator are prepared. The tool is then operated and its vibration frequency is identified within the 240–320 Hz search window. A hand-side direction is selected for control, and the actuator-to-hand secondary path is identified at the same tool-vibration frequency. These steps provide the frequency, gain and phase information required before adaptive control begins.

Once this information is available, the FxLMS controller is initialised and the system enters continuous adaptive operation. The tool-side sensor provides the live reference vibration, describing the disturbance produced by the tool, while the hand-side sensor measures the vibration remaining at the hand. The controller combines these measurements with the identified secondary-path information and updates the adaptive weight that determines the actuator command.

The updated weight changes both the amplitude and phase of the sinusoidal counter-vibration generated by the NCO. This command is amplified and applied to the exciter, and the resulting mechanical vibration reaches the hand through the secondary path. The hand-side sensor then measures the new remaining vibration, which is fed back to the controller for the next update. The sequence therefore forms a continuous closed loop in which the actuator response is progressively adjusted according to the vibration measured at the hand.

Figure 30 brings together the developed sensing, identification and adaptive-control functions. Frequency identification establishes which vibration component is to be controlled, secondary-path identification describes how the actuator response reaches the hand, and FxLMS uses this information together with the two sensor measurements to determine how the counter-vibration should be adjusted during operation.

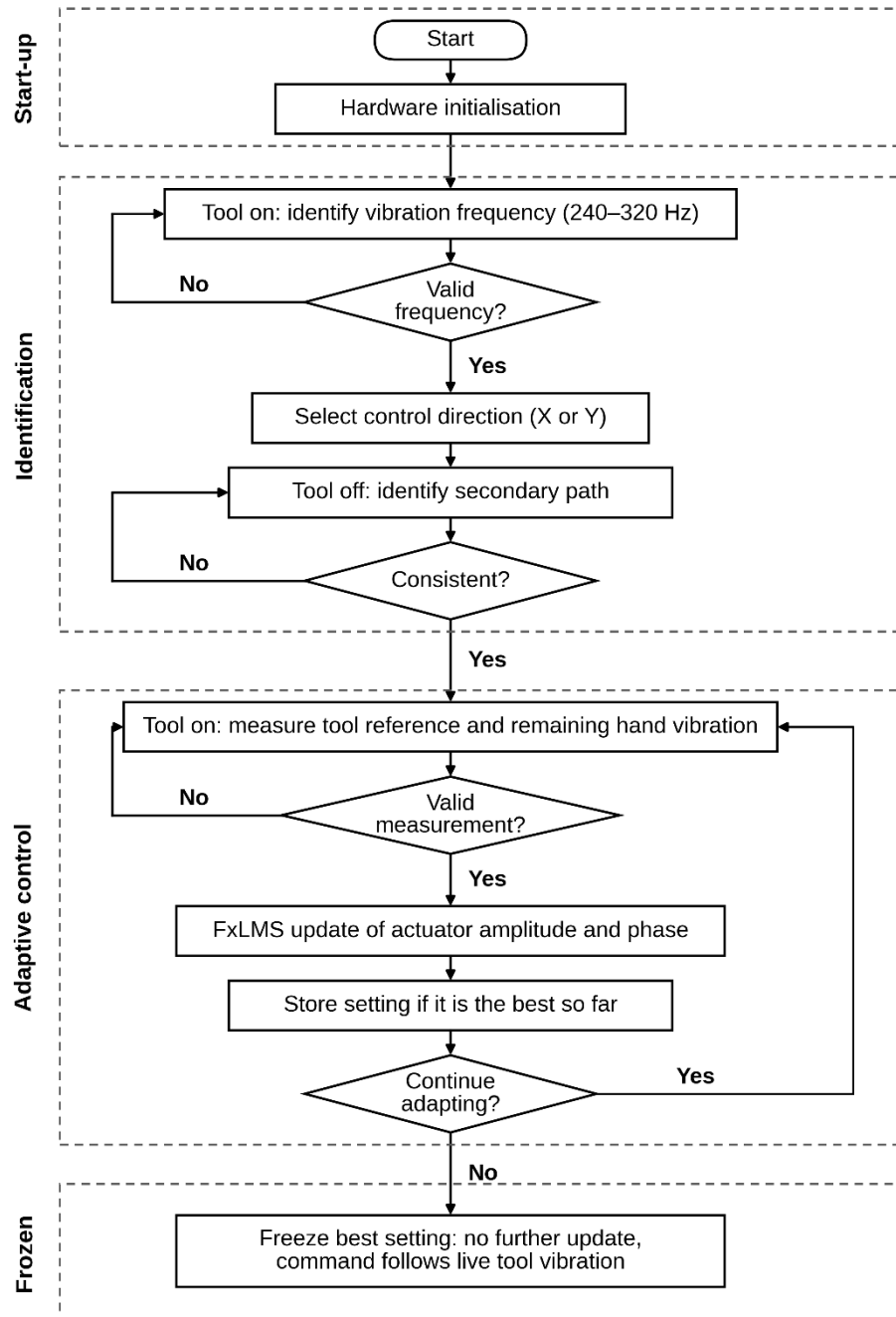


Figure 30: Control logic of the final vibration-control system.

Design decision: The final control system was therefore implemented as a sequential preparation stage followed by continuous adaptive control: initialise the hardware → identify the tool vibration frequency → identify the secondary path → initialise FxLMS → measure the live tool and hand vibrations → update the actuator amplitude and phase → repeat while the system is operating.

3.2.10 Hardware Integration and Final Prototype Configuration

The final stage of development was the integration of the selected components into the hardware configuration used for experimental evaluation.

The final system was divided into two parts: the wearable mechanical interface and the external control and power electronics. Only the components that required direct mechanical interaction with the tool or hand were mounted on the user. The tool-side ADXL345 was fixed to the tool grip, the hand-side ADXL345 was attached to the back of the hand, and the Visaton EX 45 S exciter was positioned at the wrist. The ESP32, amplifier and supporting power electronics remained externally mounted and were connected to the wearable components by cables. This arrangement kept the hand-mounted mass low while preserving the sensor and actuator locations required by the control system.

Figure 31 shows the final hardware connection used for the prototype.

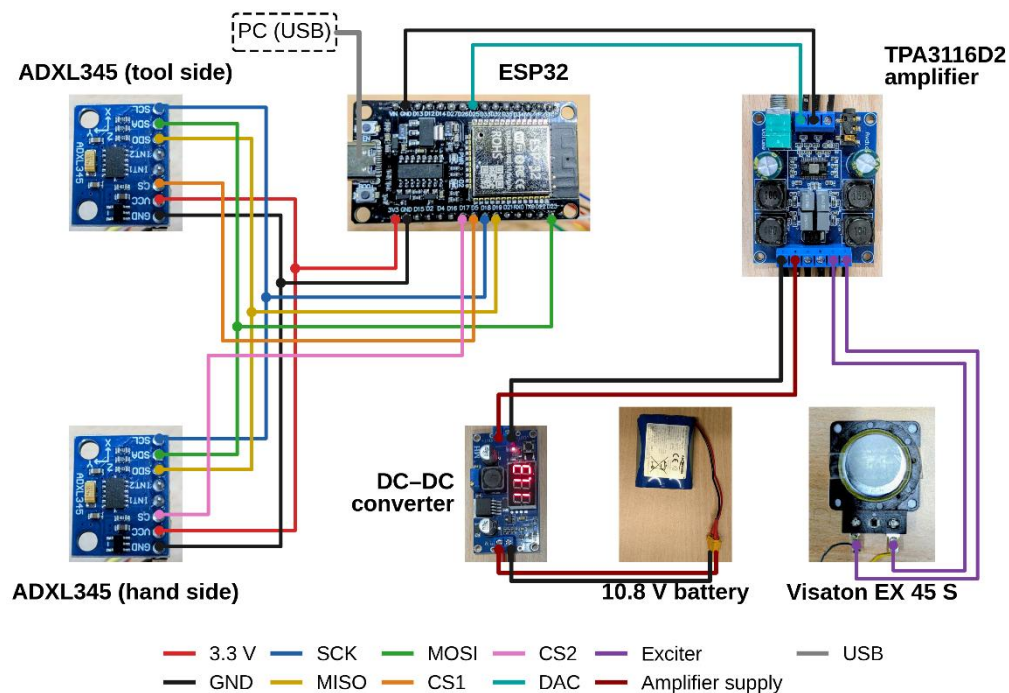


Figure 31: Wiring diagram of the final prototype.

Table 6: Final prototype components and their functional justification.

Function	Final component	Reason retained
Tool-side vibration measurement	ADXL345	Compact triaxial sensor suitable for the implemented vibration measurement
Hand-side vibration measurement	ADXL345	Provides the corresponding hand-side measurement using the same sensing platform
Embedded control	ESP32	Performs sensor acquisition, signal processing and real-time control
Mechanical actuation	Visaton EX 45 S	Produced the required controllable vibration response in the investigated frequency region
Power amplification	TPA3116D2	Provides the drive required by the exciter
Power supply	10.8 V lithium-ion battery	Provides portable power for the prototype
Voltage regulation	DC-DC converter	Provides the required regulated supply levels
Data logging and supervision	PC	Used for experimental logging and test supervision

3.3 Testing and Verification

With the development and integration of the prototype complete, the final system was evaluated through controlled experiments. The tests examined whether the FxLMS controller could learn an effective counter-vibration, whether the learned controller setting remained effective without further adaptation, whether the vibration changed consistently when active control was switched OFF and ON, and how control in one direction affected the remaining vibration directions.

Experimental Protocol

Equipment: The hand-held hot-air blower was used as the vibration source. The tool-side ADXL345 was attached to the tool grip, the hand-side ADXL345 was fixed to the back of the hand, and the Visaton EX 45 S exciter was mounted at the wrist. The ESP32 executed the real-time measurement and control functions, while the connected PC was used for test supervision and data logging.

Common test conditions: Within each comparison, the sensor positions, exciter attachment and wearing configuration were kept unchanged. Before active control was started, the current tool-vibration frequency and corresponding actuator-to-hand secondary path were identified. One hand-side direction was controlled at a time, while all three acceleration directions were recorded simultaneously.

Activities: Four evaluations were performed. First, the development of vibration reduction during FxLMS adaptation was recorded. The learned controller setting was then tested without further adaptation. Active control was subsequently switched between OFF and ON states to compare the vibration directly under both conditions. Finally, the recorded

three-axis data were examined to determine how control of one direction affected the complete hand-side vibration response.

3.3.1 Adaptive FxLMS Control Behaviour

The first evaluation examined whether the FxLMS controller could progressively develop a counter-vibration that reduced the vibration measured in the selected hand-side direction.

Method of Analysis: Before each test, the current tool-vibration frequency and corresponding secondary path were identified. One hand-side direction was then selected for control, and its vibration amplitude was measured immediately before adaptation to establish the uncontrolled reference.

The FxLMS controller was subsequently enabled for 20 s while the tool continued operating. During this period, the tool-side sensor provided the live vibration reference, and the hand-side sensor measured the vibration remaining at the identified control frequency. The hand-side vibration amplitude was recorded throughout the complete 20 s interval so that the development of the control effect could be evaluated over time.

Tests were performed separately for X- and Y-direction control using the same evaluation procedure.

Key Performance Metric: Controller performance during adaptation was expressed as the percentage change in the selected hand-side vibration relative to its value immediately before active control:

$$R(t) = \frac{A_0 - A(t)}{A_0} \times 100\% \quad (2)$$

where A_0 is the vibration amplitude measured immediately before adaptation and $A(t)$ is the amplitude measured during adaptation at time t .

A positive value of $R(t)$ indicates vibration reduction, while a negative value indicates that the vibration increased relative to the uncontrolled condition. Plotting $R(t)$ over the 20 s learning interval therefore shows how the vibration-reduction effect develops as the controller adapts.

3.3.2 Frozen Controller Validation

After the controller had developed the counter-vibration during learning, the next evaluation examined whether the learned state remained effective once adaptation stopped.

Method of Analysis: Each validation run first included the same 20 s adaptive period. After learning, the retained FxLMS weight was frozen and no further adaptive updates were applied.

Freezing the controller weight did not stop the control action. The tool-side sensor continued to provide the live vibration reference, and the frozen weight continued to define the magnitude and phase relationship used to generate the actuator command. The test therefore evaluated the learned control relationship independently of further adaptation.

For repeatability, the frozen controller was evaluated over a 10 s interval in independent X- and Y-direction tests. The mean vibration amplitude during this interval was compared with the corresponding uncontrolled reference amplitude measured before active control.

A second evaluation examined whether the frozen control effect persisted over a longer period. Two X-direction runs were therefore continued for 60 s with the controller weight held fixed. The complete 60 s record was divided into six consecutive 10 s intervals so that changes in vibration reduction over time could be examined rather than using only one average value.

Key Performance Metrics: Frozen-controller performance was expressed as the percentage reduction in the selected hand-side vibration:

$$R = \frac{A_{\text{OFF}} - A_{\text{CTRL}}}{A_{\text{OFF}}} \times 100\% \quad (3)$$

where A_{OFF} is the vibration amplitude measured before active control and A_{CTRL} is the mean vibration amplitude measured during the frozen-control interval.

For the 60 s tests, the same calculation was applied to each consecutive 10 s interval:

$$R_m = \frac{A_{\text{OFF}} - A_m}{A_{\text{OFF}}} \times 100\% \quad (4)$$

where A_m is the mean vibration amplitude in interval m .

The first metric evaluates whether the learned controller setting can reproduce vibration reduction after adaptation has stopped. The interval-based metric shows whether this reduction is maintained throughout the longer frozen-control period.

3.3.3 Vibration Response with and without Active Control

The frozen-controller tests evaluate the reduction achieved over a continuous control period. A separate switching test was used to determine whether the vibration measured at the hand changed consistently when the same control action was removed and reapplied within a single test.

Method of Analysis: The test followed an A–B–A–B sequence while the tool remained continuously operating. The controller was first switched OFF for 5 s (A_1), then switched ON for 10 s (B_1), switched OFF again for 5 s (A_2), and finally switched ON for a further 10 s (B_2).

During both ON periods, the previously learned FxLMS weight was held fixed. The tool-side reference remained active and continued to generate the actuator command using

this frozen weight. The sensor positions, exciter attachment and grip condition were kept unchanged throughout the complete switching sequence.

The vibration amplitude in the selected hand-side direction was calculated separately for each of the four intervals. The two OFF intervals were then combined to obtain the mean uncontrolled amplitude, while the two ON intervals were combined to obtain the corresponding mean amplitude with active control. Comparing the repeated OFF and ON states also showed whether the same change in vibration was reproduced after the intermediate switching event.

Key Performance Metric: The vibration reduction produced by active control was calculated from the mean OFF and ON amplitudes as:

$$R_{ABAB} = \frac{A_{OFF} - A_{ON}}{A_{OFF}} \times 100\% \quad (5)$$

where A_{OFF} is the mean selected-direction vibration amplitude from the two OFF intervals and A_{ON} is the mean amplitude from the two ON intervals.

A positive value indicates that the vibration was lower when active control was enabled. The repeated A–B–A–B sequence therefore provides a direct within-test comparison of the hand-side vibration with and without the counter-vibration under the same experimental arrangement.

3.3.4 Cross-Axis and Overall Vibration Response

The previous evaluations considered the vibration reduction in the selected control direction. The final evaluation examined how this control action affected the other hand-side directions and the combined three-axis vibration response.

Method of Analysis: The analysis reused the frozen-controller measurements. For each test, the vibration amplitudes in the X-, Y- and Z-directions were obtained for the uncontrolled reference condition and for the corresponding frozen-control interval.

The change in each direction was calculated separately so that vibration reduction in the controlled direction could be compared with any increase or decrease occurring in the remaining directions. The three measured amplitudes were also combined into a resultant amplitude to assess the overall vibration response at the hand-side sensor.

Key Performance Metrics: The percentage change in each measured direction was calculated as:

$$R_k = \frac{A_{OFF,k} - A_{CTRL,k}}{A_{OFF,k}} \times 100\% \quad (6)$$

where $A_{OFF,k}$ is the vibration amplitude in direction k without active control and $A_{CTRL,k}$ is the corresponding amplitude with the frozen controller active.

A positive value of R_k indicates a reduction in that direction, while a negative value indicates an increase. The resultant vibration amplitude was calculated from the three measured directions as:

$$A_{\text{res}} = \sqrt{A_X^2 + A_Y^2 + A_Z^2} \quad (7)$$

The corresponding change in resultant amplitude was then calculated as:

$$R_{\text{res}} = \frac{A_{\text{res,OFF}} - A_{\text{res,CTRL}}}{A_{\text{res,OFF}}} \times 100\% \quad (8)$$

where $A_{\text{res,OFF}}$ and $A_{\text{res,CTRL}}$ are the resultant amplitudes without and with active control, respectively.

These metrics allow the selected-direction reduction to be considered together with any redistribution of vibration into the remaining directions and the resulting change in the overall three-axis response.

3.3.5 Ergonomics and Perceived Efficacy

The preceding evaluations quantify the vibration-control performance of the prototype. Since the system is worn directly on the hand and wrist during tool operation, a final evaluation was included to assess how the prototype was experienced by the user.

Method of Analysis: After completing the vibration-control experiments, the participants completed a short, structured questionnaire covering the main human-use aspects of the prototype. The questions considered wearing comfort, attachment stability, freedom to grip and operate the tool, interference from the wearable components, perceived vibration reduction and overall confidence while using the system.

Each statement was rated using a 7-point Likert scale, where 1 represented the most negative response, 4 represented a neutral response and 7 represented the most positive response. The scale provides sufficient distinction between responses while remaining simple for the participant to use immediately after testing.

Evaluation Criteria: The questionnaire was organised around three aspects of the wearable system:

- **Ergonomics:** comfort of the hand- and wrist-mounted components and freedom of hand movement.
- **Usability:** ease of wearing the prototype, stability of the attachments and ability to grip and operate the tool.
- **Perceived efficacy:** whether the participant could perceive a reduction in vibration when active control was operating.

The responses were evaluated descriptively and were used to complement the measured vibration results rather than as a separate quantitative performance measure.

4 Results and Discussion

This chapter presents and discusses the experimental performance of the final active vibration-control system. The analysis considers the vibration reduction achieved in the controlled direction together with its repeatability, persistence, cross-axis effects and wearable usability, allowing the completed prototype to be assessed against the original design requirements.

4.1 Adaptive FxLMS Control Behaviour

This section evaluates how the FxLMS controller develops vibration reduction during the 20 s learning period. The behaviour is examined for control in both the X and Y directions to show how the reduction changes as the controller adjusts the actuator response.

4.1.1 Results

Figure 32 shows two X-direction and two Y-direction learning traces. The percentage reduction is calculated by comparing the vibration during active control with the vibration measured immediately before control begins. Positive values indicate that the vibration is reduced, while negative values indicate a temporary increase.

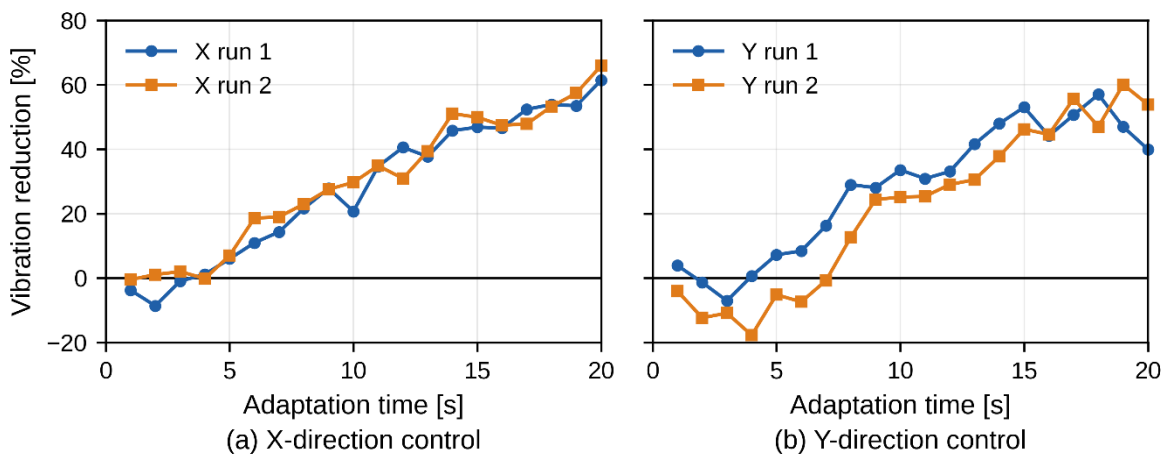


Figure 32: Vibration reduction during FxLMS learning with (a) X-direction and (b) Y-direction control.

For X-direction control, both tests begin with little or no reduction. From approximately 5 s onward, the reduction increases progressively. During the later part of the learning period, both tests exceed 50% reduction and reach 61.4% and 65.9% at the end of the 20 s interval. The two tests follow a similar overall development.

The Y-direction tests show the same general trend, although with larger variations. During the first few seconds, the measured vibration temporarily increases while the controller is still adapting. The reduction then develops progressively, and both tests exceed 50%

during the later part of the learning period. Maximum reductions of approximately 57.0% and 60.1% are reached.

4.1.2 Discussion

The results show that the controller gradually develops an actuator response that opposes the vibration produced by the tool. At the beginning of learning, the actuator output is not yet properly adjusted in magnitude and timing, so only a small reduction is obtained and the vibration can temporarily increase. As the controller continues to adapt, the actuator response becomes more effective and the measured vibration decreases.

The X-direction tests show a smoother development, while the Y-direction tests contain larger short-term variations. Nevertheless, both directions reach substantial vibration reduction within the same 20 s learning period. This shows that the adaptive approach can learn an effective counteracting response in more than one direction of the developed system.

The Z direction is not evaluated as a separate control direction because the vibration component around the investigated control frequency is considerably weaker and less consistently measurable than in X and Y. The adaptive-control evaluation therefore focuses on the two directions that provide the clearest and most repeatable vibration components for control.

The learning results establish that substantial selected-direction attenuation can develop during adaptation. The next question is whether the learned weight remains effective after adaptation is disabled.

4.2 Frozen Controller Validation

After each learning period, the learned controller weight is frozen and evaluated without further adaptation. The tests examine whether the resulting vibration reduction can be reproduced and maintained over a longer measurement period.

4.2.1 Results

Figure 33 shows the vibration reduction measured during repeated frozen-controller tests in the X and Y directions. Each bar represents an independent test in which the controller first learns and is then evaluated again with learning disabled.

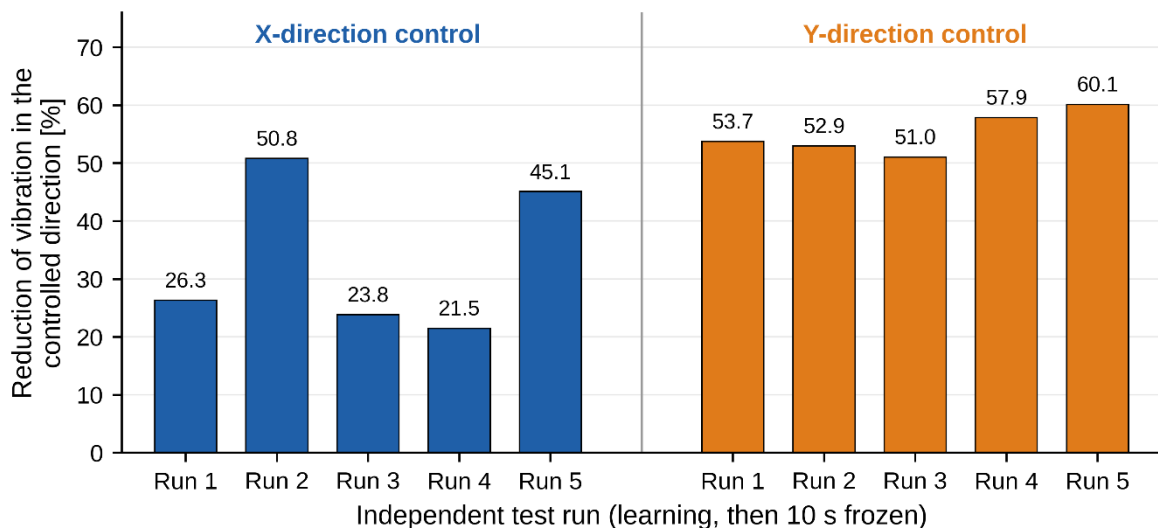


Figure 33: Vibration reduction during repeated frozen-controller tests in the X and Y directions.

For X-direction control, all five tests show a positive reduction in vibration. The measured reductions range from 21.5% to 50.8%, with a mean value of 33.5%. This shows that the learned controller repeatedly reduces vibration in the X direction, although the reduction level varies from run to run.

For Y-direction control, all five retained tests also show positive vibration reduction. The measured reductions range from 51.0% to 60.1%, with a mean value of 55.1%. Compared with the X-direction tests, the Y-direction results are more closely grouped and remain above 50% in all five runs.

Figure 34 shows the longer-duration validation for two X-direction tests continued for 60 s with learning stopped. The results are presented in six consecutive 10 s intervals.

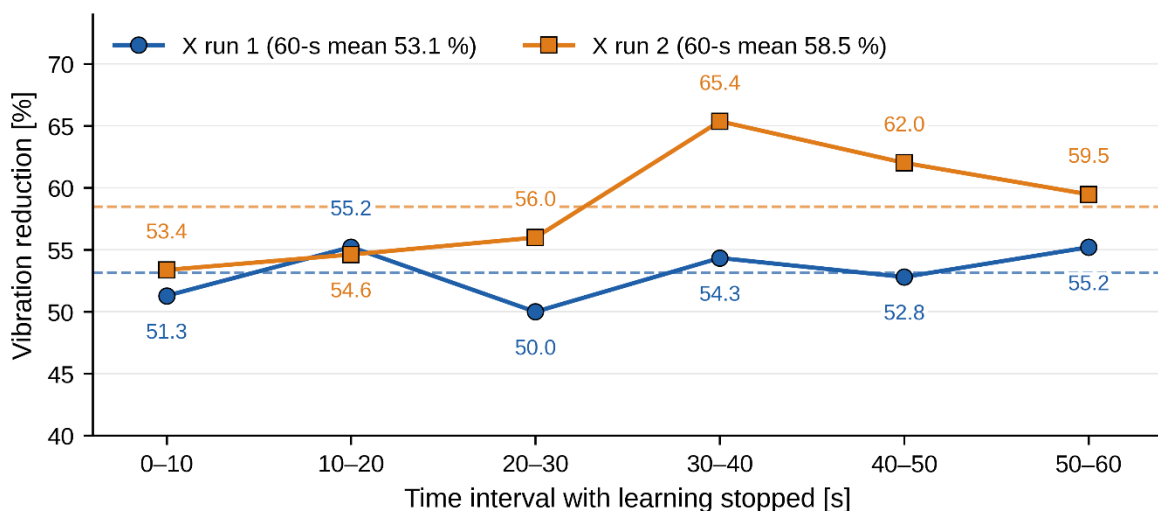


Figure 34: Vibration reduction during two 60 s frozen-controller tests in the X direction.

For Run 1, the average vibration reduction over the complete 60 s period is 53.1%. The six intervals remain between 50.0% and 55.2%. For Run 2, the average reduction is 58.5%, and the six intervals remain between 53.4% and 65.4%. In both cases, the vibration reduction remains positive throughout the full 60 s test.

Figure 35 places the peak reduction observed during adaptation alongside the mean reductions from the subsequent 10 s and 60 s frozen-weight intervals. Because the learning value is a peak while the frozen values are interval means, these values are not directly equivalent performance statistics; the comparison is used only to show that substantial attenuation persists after adaptation stops.

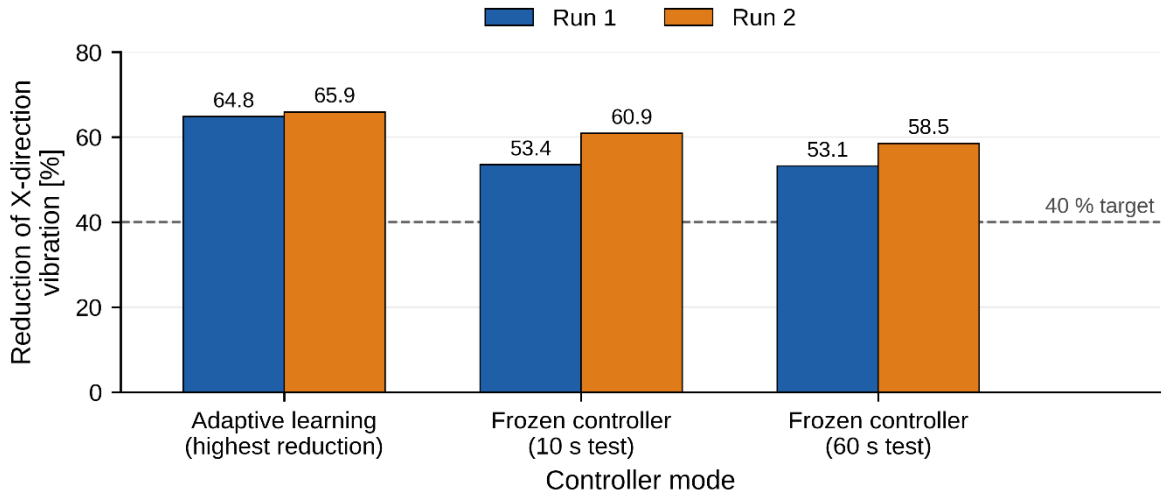


Figure 35: Comparison of peak adaptive-learning reduction with mean 10 s and 60 s frozen-weight reduction in two X-direction tests.

In both runs, the peak reduction during adaptation is 64.8 % and 65.9 %, while the corresponding mean 10 s frozen-weight reductions are 53.4 % and 60.9 % and the mean 60 s reductions are 53.1 % and 58.5 %. The mean frozen-weight reductions therefore remain above 50 % in both runs after adaptation is disabled.

4.2.2 Discussion

The frozen-controller results show that the vibration reduction is not limited to the learning phase alone. In the repeated X- and Y-direction tests, the controller continues to reduce vibration after learning is disabled. This means that the learned controller state remains effective when it is tested again independently.

The 60 s X-direction results further show that the reduction is not only a short-lived or instantaneous effect. In both runs, the reduction remains positive over the full 60 s measurement and does not show a progressive decline. This indicates that the learned controller state remains usable over a longer time interval under the investigated test condition.

An important point is that only the learned controller weight is frozen. The vibration from the tool is still measured continuously, so the controller continues to respond to the current vibration while keeping the learned relationship unchanged. In other words, the controller does not replay one fixed stored waveform. It still follows the live reference signal, but it no longer updates what it has learned. This explains why the frozen controller can remain effective even if the phase of the incoming tool vibration changes during the test.

The selected durations of 20 s for learning and 60 s for the longer frozen test are fixed experimental windows used for consistent evaluation. They are not special values required by the FxLMS algorithm itself. The 20 s interval is long enough to show the development of controller learning, while the 60 s interval provides a practical check of whether the reduction continues after learning has stopped.

Overall, the frozen-controller results confirm two main findings. First, the achieved vibration reduction is reproducible across repeated tests in both the X and Y directions.

Second, the learned controller state remains effective over a longer continuous measurement without further adaptation.

A direct OFF–ON comparison provides an additional test of whether applying and removing the learned control action produces a corresponding change in the measured vibration.

4.3 Vibration Response with and without Active Control

This section compares the vibration measured with the controller switched OFF and ON while the vibration source continues operating. Each test consists of four consecutive phases: 5 s OFF (A_1), 10 s ON (B_1), 5 s OFF (A_2), and 10 s ON (B_2). The controller uses the previously learned setting throughout the sequence, and no further adaptation takes place during the ON periods. Repeating both operating states within the same measurement allows the effect of active control to be compared directly.

4.3.1 Results

Figure 36 summarizes the three retained successful switching tests. To make the effect of controller activation directly comparable despite the different initial vibration amplitudes, Figure 36(a) normalizes the mean vibration measured during the OFF periods to 100% for each test. Figure 36(b) presents the corresponding measured vibration amplitudes in m/s^2 .

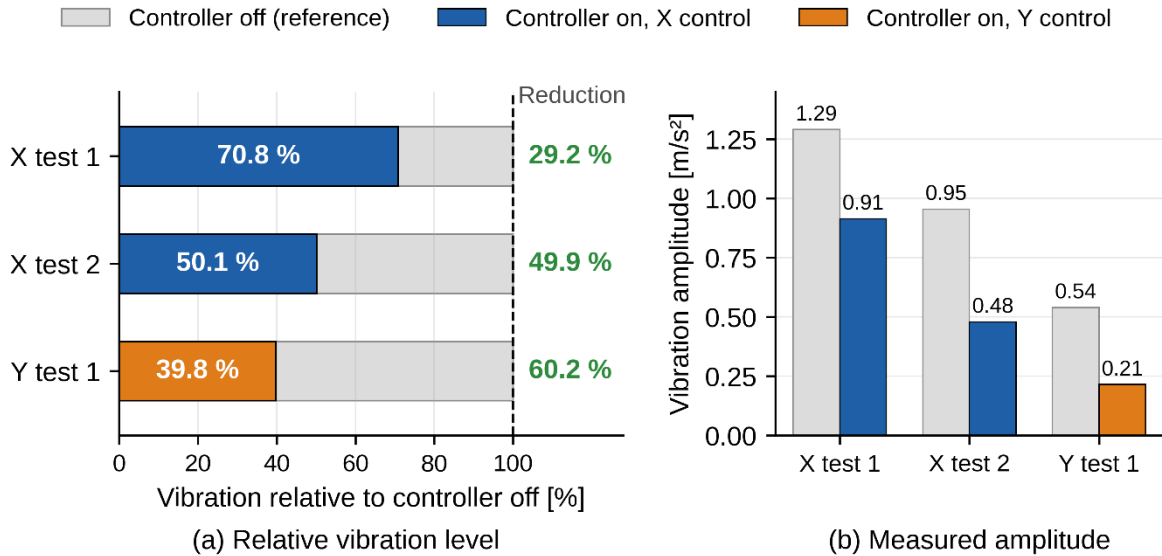


Figure 36: OFF-ON switching validation: (a) vibration relative to the control-OFF reference; (b) corresponding mean selected-direction vibration amplitudes.

In the first X-direction test, the mean vibration decreases from 1.290 m/s² with the controller OFF to 0.914 m/s² with the controller ON. The remaining vibration is therefore 70.9% of the OFF condition, corresponding to a reduction of 29.15%.

The second X-direction test shows a larger change, from 0.955 m/s² to 0.478 m/s². The vibration during active control is 50.1% of the OFF condition, corresponding to a reduction of 49.92%.

The Y-direction test shows the largest reduction. The measured vibration decreases from 0.539 m/s² to 0.214 m/s², leaving 39.8% of the vibration measured with the controller OFF. This corresponds to a reduction of 60.22%.

Thus, all three retained switching tests show the same response: activating the controller reduces the vibration measured in the direction being controlled.

4.3.2 Discussion

The switching tests provide a particularly intuitive comparison of the developed system with and without active vibration control. Because the tool continues operating while the controller alternates between OFF and ON states, both conditions are evaluated within the same test rather than through separate measurements. The repeated sequence also shows that the reduction is reproduced when active control is applied again after an intermediate OFF period.

The magnitude of the reduction differs between the tests, ranging from approximately 29% to 60%. This variation is consistent with the run-to-run variability observed in the frozen-controller tests and reflects the sensitivity of the active system to the mechanical coupling between the actuator, wearable structure and hand. Nevertheless, the direction of the response remains consistent across both investigated control directions: the vibration in the controlled direction is lower with active control than without it.

During these measurements, only the learned controller weight remains fixed. The vibration from the tool continues to provide the live reference used to generate the actuator command. The controller therefore responds to the current disturbance rather than replaying a previously stored vibration signal.

The OFF–ON measurements therefore provide the clearest direct comparison of the developed system's vibration-control effect. The remaining question is whether reducing the targeted direction also alters vibration along the other axes.

4.4 Cross-Axis and Overall Vibration Response

Because the actuator force is transmitted through the wearable structure in three dimensions, reducing one direction can also change vibration along the remaining axes. The frozen-controller data were therefore examined in X, Y and Z together with the three-axis resultant.

4.4.1 Results

Figure 37 shows the percentage change in vibration during frozen active control for five X-direction and five retained Y-direction tests. Positive values indicate vibration reduction, while negative values indicate an increase relative to the corresponding uncontrolled condition. The outlined column identifies the direction targeted by the controller.

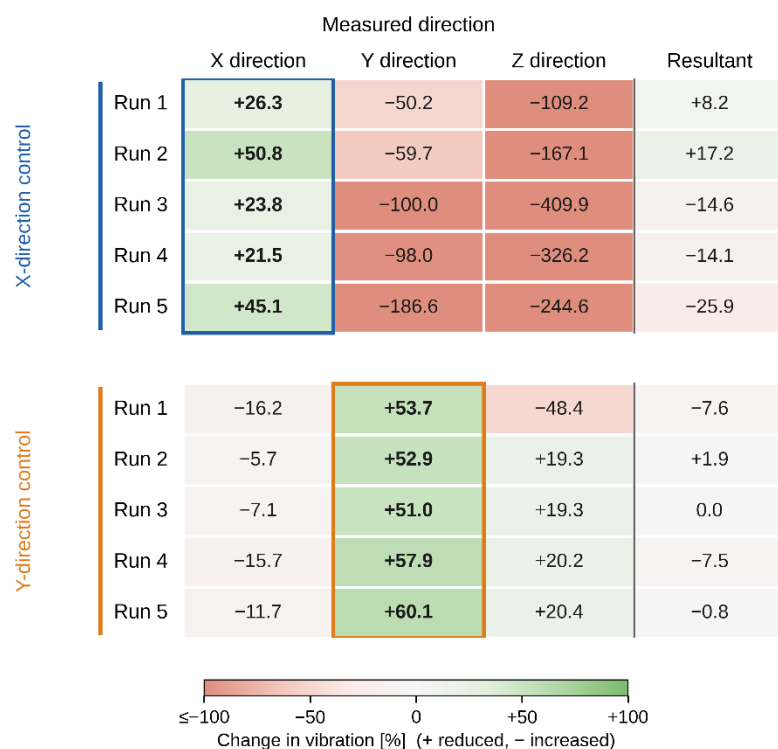


Figure 37: Cross-axis and three-axis resultant vibration response during frozen control.

For X-direction control, the targeted X vibration decreases in all five tests, with reductions between 21.5% and 50.8%. However, vibration increases in both uncontrolled directions. The Y component increases by 50.2–186.6%, while the Z component increases by 109.2–

409.9%. The resultant response is therefore mixed: it decreases by 8.2% and 17.2% in two tests but increases by 14.1–25.9% in the remaining three.

The very large percentage changes in the Z direction should be interpreted with care. The uncontrolled Z vibration is comparatively small, so a relatively modest absolute change in vibration can produce a much larger percentage change. The percentage values therefore describe the change relative to the initial Z level and should not be interpreted as equally large absolute vibration amplitudes.

For Y-direction control, the targeted Y vibration decreases consistently by 51.0–60.1%. The X component increases moderately by 5.7–16.2% in all five tests. The Z component decreases by approximately 19–20% in four tests, while one test shows an increase of 48.4%. In contrast to the large reduction achieved in Y, the resultant remains close to its uncontrolled level, ranging from a 1.9% reduction to a 7.6% increase, with one test showing essentially no change.

Overall, the frozen measurements show two distinct response patterns. X-direction control produces repeatable X attenuation but substantial cross-axis amplification, particularly in Z, with the resultant improving in two tests and increasing in three. Y-direction control produces strong and repeatable Y attenuation, while the resultant changes only moderately and remains approximately around the uncontrolled level.

4.4.2 Discussion

The results show that effective reduction in the controlled direction does not necessarily produce an equivalent reduction in the total three-axis vibration. The present controller minimizes vibration along one direction at a time, while the other measured directions are not included directly in its control objective. The actuator force can therefore reduce vibration along the targeted axis while simultaneously changing the response along the remaining axes.

This behaviour is particularly evident during X direction control. Although X vibration is reduced in every frozen test, the Y and Z components increase in all five cases. As a result, the reduction obtained along X translates into lower resultant vibration in only two tests. The response during Y-direction control is more favourable from a cross-axis perspective: the controlled Y component is reduced by more than 50% in every retained test, and Z is also reduced in four of the five measurements. Nevertheless, the resultant remains close to the uncontrolled condition because the reductions and increases occurring along the different axes partly offset one another.

These results define an important distinction in the performance of the developed system. The experiments demonstrate repeatable narrowband attenuation in the direction targeted by the controller, but they do not demonstrate simultaneous attenuation of all three vibration components. This behaviour is consistent with the present single-actuator, single-direction control configuration, in which the mechanical coupling between the actuator, wearable structure and hand determines how the generated counter-vibration is distributed spatially.

The cross-axis response therefore represents a limitation of the current proof-of-concept architecture rather than a contradiction of the experimentally demonstrated selected-direction reductions. Extending the system towards overall hand–arm vibration reduction would require the control objective and actuator arrangement to account explicitly for vibration in multiple directions.

4.5 Ergonomics and Perceived Efficacy

Technical performance alone does not establish how the prototype is experienced during tool use. Three participants therefore evaluated its ergonomics, usability and perceived vibration reduction.

4.5.1 Results

The results from the 7-point Likert scale questionnaire, summarizing the ratings for ergonomics, usability and perceived efficacy of the developed prototype, are presented in Table 7.

The scale used: 1 = Strongly Disagree, 2 = Disagree, 3 = Slightly Disagree, 4 = Neutral, 5 = Slightly Agree, 6 = Agree, 7 = Strongly Agree

Table 7: Participant ratings of prototype ergonomics, usability and perceived efficacy.

No.	Statement	P1	P2	P3	Median
1	The prototype is easy to put on and remove.	4	5	5	5
2	The hand- and wrist-mounted components are comfortable during testing.	2	3	4	3
3	The straps hold the sensor and exciter securely during tool operation.	6	6	6	6
4	The prototype allows me to grip and operate the tool normally.	4	6	6	6
5	The wearable components do not noticeably restrict my hand or wrist movement.	3	2	3	3
6	The cables and external electronics do not significantly interfere with tool operation.	3	3	2	3
7	I can perceive a reduction in vibration when active control is operating.	5	5	6	5
8	I feel comfortable and confident operating the tool while wearing the prototype.	4	5	5	5

4.5.2 Discussion

The subjective evaluation shows that the current prototype performs well in the aspects required to maintain the sensing and actuation arrangement during testing, while the main limitations arise from the physical integration of the wearable components and external electronics.

The attachment system received the strongest response, with all three participants rating its stability at 6. The adjustable straps therefore provided sufficiently secure fixation of the

hand-side accelerometer and wrist-mounted exciter during tool operation. This is particularly important for the developed control system because movement of either component changes the mechanical connection through which the counter-vibration is measured and transmitted.

Putting on and removing the prototype received a median rating of 5. The relatively simple strap-based attachment therefore made the system reasonably easy to wear without requiring a complex fitting procedure. Normal tool gripping received a median rating of 6. Because the accelerometer is located on the back of the hand and the exciter at the wrist, the palm and fingers remain free, allowing the tool to be held and operated normally. The neutral rating of one participant is consistent with the restriction caused by the connecting cables. This supports the original wearability requirement that the palm and fingers remain free for normal gripping.

The same arrangement was less successful in terms of comfort. The hand- and wrist-mounted components received a median comfort rating of 3. In the present prototype, the hand-side ADXL345 is strapped directly against the back of the hand and the exciter is held against the wrist. Although this arrangement provides the mechanical contact required for vibration measurement and actuation, direct contact with the components and strap pressure can become uncomfortable during longer periods of use. Future development should therefore incorporate dedicated sensor and actuator interfaces with suitable padding while maintaining sufficiently rigid mechanical coupling.

Freedom of hand and wrist movement also received a median rating of 3. The palm and fingers remain available for gripping, but the wrist-mounted exciter, straps and connected wiring limit unrestricted movement around the wrist. The prototype therefore allows the required tool operation, but it does not yet provide the mobility expected from a fully integrated wearable system.

The clearest usability limitation was the external electrical implementation. The statement that the cables and external electronics did not interfere with operation received a median rating of 3. The wearable sensors and actuator are connected to the bench-mounted controller and supporting electronics through cables and jumper wires. As a result, the tool cannot be moved freely over a large working area and the user must remain aware of the electrical connections during operation. This configuration was sufficient for controlled experimental testing, but miniaturisation of the electronics, improved cable routing and integration of the control and power hardware into a wearable unit are required before practical mobile use can be considered.

Perceived vibration reduction received a positive median rating of 5, with all three participants assigning ratings between 5 and 6. The participants could therefore perceive a reduction in vibration when active control was operating. This subjective response is consistent with the measured reductions obtained during the frozen-controller and OFF-ON tests.

However, the perceived effect should not be expected to correspond directly to the selected-axis percentage reduction. The present system controls only one vibration direction at a time, while vibration in the remaining directions is still experienced by the

user and can change because single-axis control also alters the coupled mechanical response. A strong measured reduction in the controlled direction may therefore produce a more moderate change in the overall vibration perceived by the user. This provides a plausible explanation for the median perceived-efficacy rating of 5 rather than the highest values on the scale.

Overall comfort and confidence during tool operation received a median rating of 5. The main weaknesses are therefore associated with the current physical implementation: direct component contact, restricted wrist movement and the tethered external electronics.

The evaluation involved only three participants and therefore represents an initial usability assessment rather than a general evaluation of user acceptance. One response was provided by the developer of the prototype and may therefore be influenced by greater familiarity with the system. In addition, the participants knew when active control was operating, which may have influenced the perceived-efficacy ratings. Nevertheless, the feedback identifies a consistent set of practical design priorities for the next prototype iteration: improved hand and wrist interfaces, greater freedom of movement and integration of the external electronics into a more compact wearable configuration.

4.6 Overall System Assessment

The completed prototype was assessed against the original design requirements. Table 8 shows how each requirement was addressed by the experimental evidence. The requirements were therefore addressed at both the technical and wearable levels. The 40% vibration-reduction target was exceeded in all retained Y-direction frozen tests and in selected X-direction validation runs, including both 60 s tests. The X-direction repeatability remained more variable. The measurement requirement was fulfilled by simultaneous three-axis sensing at the tool and hand, while the active-control requirement was realised through automatic adjustment of the counter-vibration at the identified operating frequency. Real-time operation was achieved by maintaining measurement, signal processing and actuator control while the tool continued running. The wearable arrangement kept the palm and fingers free for gripping, which was also reflected in the positive user rating for normal tool operation. Finally, the strap-based mounting provided sufficiently stable sensor and actuator attachment during the experiments, supported by the consistently high attachment-stability rating.

Table 8: Assessment of the final prototype against the design requirements.

Requirement	Final evidence	Assessment
Vibration cancellation	>50% reduction in repeated Y tests and X tests	Demonstrated, with X-direction variability
Vibration measurement	Tool and hand vibration measured in X, Y and Z	Achieved
Active control	Frequency, amplitude and phase adjusted during adaptive control	Achieved
Real-time operation	Measurement and control operated continuously while the tool was running	Achieved
Wearability	Palm and fingers remained free; grip rating 6/7	Achieved for the test configuration
Mechanical integration	Sensor and exciter securely attached; stability rating 6/7	Achieved for controlled testing

The 40% performance target also provides a useful basis for placing the developed system in the context of published vibration-mitigation approaches. Table 9 compares the present results with experimental systems reported in the literature. The values should be considered together with the corresponding control principle and experimental conditions.

Table 9: Comparison of vibration reduction with published vibration-mitigation approaches.

Study / approach	Reported reduction	Context
ISO 10819 / Hewitt et al. [ISO10819; Hew16]	$\geq 40\%$	High-frequency anti-vibration glove benchmark
Ciullo et al. [Ciu21]	40–60%	Wearable robotic system during drilling
Abdul Satar et al. [Sat21]	up to 39.7%	Active suspended handle under shaker excitation
Ahmad Mazlan and Ripin [Maz17]	91%	Active suspended handle on a die grinder
Present work	51.0–60.1% Y; 53.1–58.5% in 60 s X tests	Wearable adaptive narrowband control at the hand

Ciullo et al. provide the closest experimental comparison to the present work, reporting a 40–60% reduction in vibration transmission during human drilling tests. The reduction demonstrated by the present prototype under its stronger validated conditions lies within this range. Active suspended handles have achieved both lower and substantially higher reductions, but these systems modify the tool–handle structure directly rather than applying the counter-vibration through a wearable hand/wrist interface.

The comparison must therefore be interpreted together with the control objective. The present system reduces one dominant vibration component in one selected direction, whereas the literature values can represent frequency-weighted transmission or vibration transmissibility over different conditions.

Taken together, the experiments and the user evaluation demonstrate that active narrowband cancellation can be implemented directly at a wearable hand–wrist interface and can produce repeatable reductions above the original 40% design target under suitable mechanical conditions. The user evaluation further shows that this control effect is perceptible during operation while the prototype maintains secure attachment and normal tool gripping. The remaining challenge is no longer whether the principle can generate meaningful vibration reduction, but how to make that reduction equally repeatable across wearing conditions and effective across multiple vibration directions while improving the physical integration of the wearable system.

5 Summary and Outlook

This master's thesis develops and experimentally validates a wearable active vibration-control hand exoskeleton for reducing vibration transmitted from a powered hand tool to the hand. The developed prototype combines tool- and hand-side vibration measurement, electrodynamic actuation at the wrist and adaptive FxLMS control. The controller identifies the dominant vibration component, accounts for the actuator response at the hand and adjusts the generated counter-vibration accordingly. The final system is evaluated through controlled vibration tests and a preliminary user evaluation.

Key Findings

- Consistent frequency identification: The two accelerometers show a mean absolute frequency difference of approximately 0.006 Hz during common-source dynamic testing, and the dominant tool vibration is identified in all runs within 267.5–272.1 Hz.
- Effective vibration reduction: Repeated frozen-controller tests achieve 21.5–50.8% reduction in X and 51.0–60.1% in Y, with mean reductions of 33.5% and 55.1%, respectively.
- Persistent and reproducible reduction: Two 60 s X-direction tests maintain average reductions of 53.1% and 58.5%. OFF–ON tests produce reductions of 29.15% and 49.92% in X and 60.22% in Y, consistently showing lower vibration with active control.
- Positive wearable performance: Secure attachment and normal tool gripping both receive median ratings of 6/7, while perceived vibration reduction receives 5/7, showing that the control effect is noticeable during tool operation.

Limitations and Future Outlook

- Mechanical repeatability: The amount of vibration reduction changes when the contact between the actuator, straps and hand changes. Small differences in strap tension, wrist position or actuator contact can therefore lead to different results between repeated tests. Future designs should use a more defined and comfortable mounting system that keeps the actuator in the same mechanical condition each time, while allowing the controller to update its actuator-to-hand response automatically when the wearing condition changes.
- Multi-axis vibration control: The present prototype controls one selected direction at a time. The cross-axis measurements show that reducing one component can also change the vibration measured in the remaining directions. Future systems should therefore investigate coordinated actuation and control in multiple directions.

- Extension beyond one dominant frequency: The controller currently targets the dominant vibration component around 268–272 Hz. Future work should extend the control strategy towards multiple-frequency or broader-band cancellation so that a larger portion of tool-generated vibration can be addressed.
- Wearable integration: The sensing and actuation elements are worn on the hand and wrist, while the controller and supporting electronics remain externally connected by cables. The user evaluation identifies comfort, wrist freedom and cable interference as the main practical weaknesses. A future prototype should therefore integrate the electronics into a more compact wearable configuration and improve the hand and wrist interfaces.
- Broader experimental validation: The present study evaluates one vibration source under controlled conditions, with a preliminary usability assessment involving three participants. Further validation should include additional powered tools, longer operating periods and a larger participant group. Standardised broadband hand–arm vibration measurements would also allow the active system to be evaluated against occupational exposure quantities.

Conclusion

This thesis demonstrates the feasibility of wearable active vibration cancellation at the hand using vibration sensing, electrodynamic actuation and adaptive FxLMS control. The developed prototype achieves substantial selected-direction vibration reduction, reaching up to 60.1 %, while the control effect remains effective during continued operation and was reported as perceptible by the three questionnaire participants during tool use. The results establish a strong experimental basis for the further development of wearable active vibration-control systems. Future development should focus on more consistent mechanical attachment, multi-axis control and improved integration of the wearable hardware.

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Appendix

A. Summary of Signal-Processing and Control Parameters

The vibration-control system combines embedded acceleration measurement, narrowband frequency identification, actuator characterisation and adaptive FxLMS control. The corresponding methods and evaluation equations are described in Chapter 3; the principal implementation parameters are summarised here for reference.

Two ADXL345 accelerometers were operated at an output data rate of 1600 Hz with a measurement range of ± 16 g. Frequency identification was restricted to the 240–320 Hz search range. The initial estimate was obtained from three averaged 1024-sample FFT records, giving a frequency-bin spacing of approximately 1.56 Hz. This estimate was refined by quadrature processing over a 2560-sample record divided into ten blocks of 256 samples, corresponding to approximately 160 ms per block.

The actuator response was characterised at 268, 270 and 275 Hz. During amplitude matching, the actuator response was accepted when it lay within ± 5 % of the required target amplitude. The subsequent open-loop phase investigation covered 0° to 330° in 30° increments. The actuator command was generated at an update rate of 10 kHz.

During adaptive control, the FxLMS evaluation and update interval was approximately 160 ms. Each learning test used a 20 s adaptation period. For frozen-weight validation, the learned adaptive weight was held fixed and evaluated over 10 s intervals, while the longer persistence tests continued for 60 s and were analysed in six consecutive 10 s intervals. The OFF–ON validation followed a 5 s OFF $\rightarrow$ 10 s ON $\rightarrow$ 5 s OFF $\rightarrow$ 10 s ON sequence.

The vibration-reduction quantities used for evaluation are defined in Section 3.3. Positive reduction values indicate lower vibration than the corresponding pre-control or control-OFF reference. The three-axis resultant is used only for comparison within the present experiments and does not represent the frequency-weighted vibration total value used for occupational hand–arm vibration exposure assessment according to ISO 53491.

